



US012416696B2

(12) **United States Patent**
Stickle et al.

(10) **Patent No.:** **US 12,416,696 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **MODULAR RADIO-FREQUENCY COIL ASSEMBLIES OF A MAGNETIC RESONANCE SYSTEM**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 149 days.

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(21) Appl. No.: **18/166,820**

(22) Filed: **Feb. 9, 2023**

Primary Examiner — Gregory H Curran

(65) **Prior Publication Data**

(74) *Attorney, Agent, or Firm* — Armstrong Teasdale LLP

US 2024/0272254 A1 Aug. 15, 2024

(51) **Int. Cl.**
G01R 33/34 (2006.01)
A61B 5/055 (2006.01)

(57) **ABSTRACT**

(52) **U.S. Cl.**
CPC **G01R 33/34084** (2013.01); **A61B 5/055** (2013.01)

A radio frequency (RF) coil assembly for a magnetic resonance (MR) system is provided. The RF coil assembly includes one or more modules. A module of the one or more modules includes one or more RF coils, each RF coil including a coil loop that includes a wire conductor, the wire conductor formed into the coil loop. Each RF coil also includes a coupling electronics portion coupled with the coil loop. The module further includes an outlet coupled with coupling electronics portions of the one or more RF coils. The RF coil assembly further includes a wiring harness including module interfacing cables each configured to be electrically coupled with the outlet.

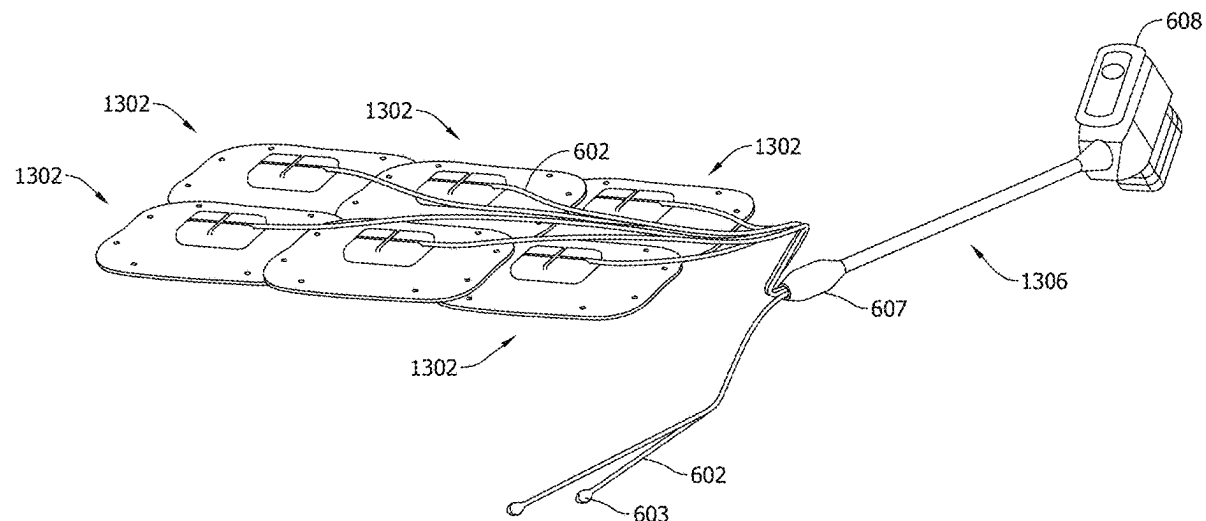
(58) **Field of Classification Search**
CPC A61B 5/055; G01R 33/34084
See application file for complete search history.

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19 Claims, 16 Drawing Sheets



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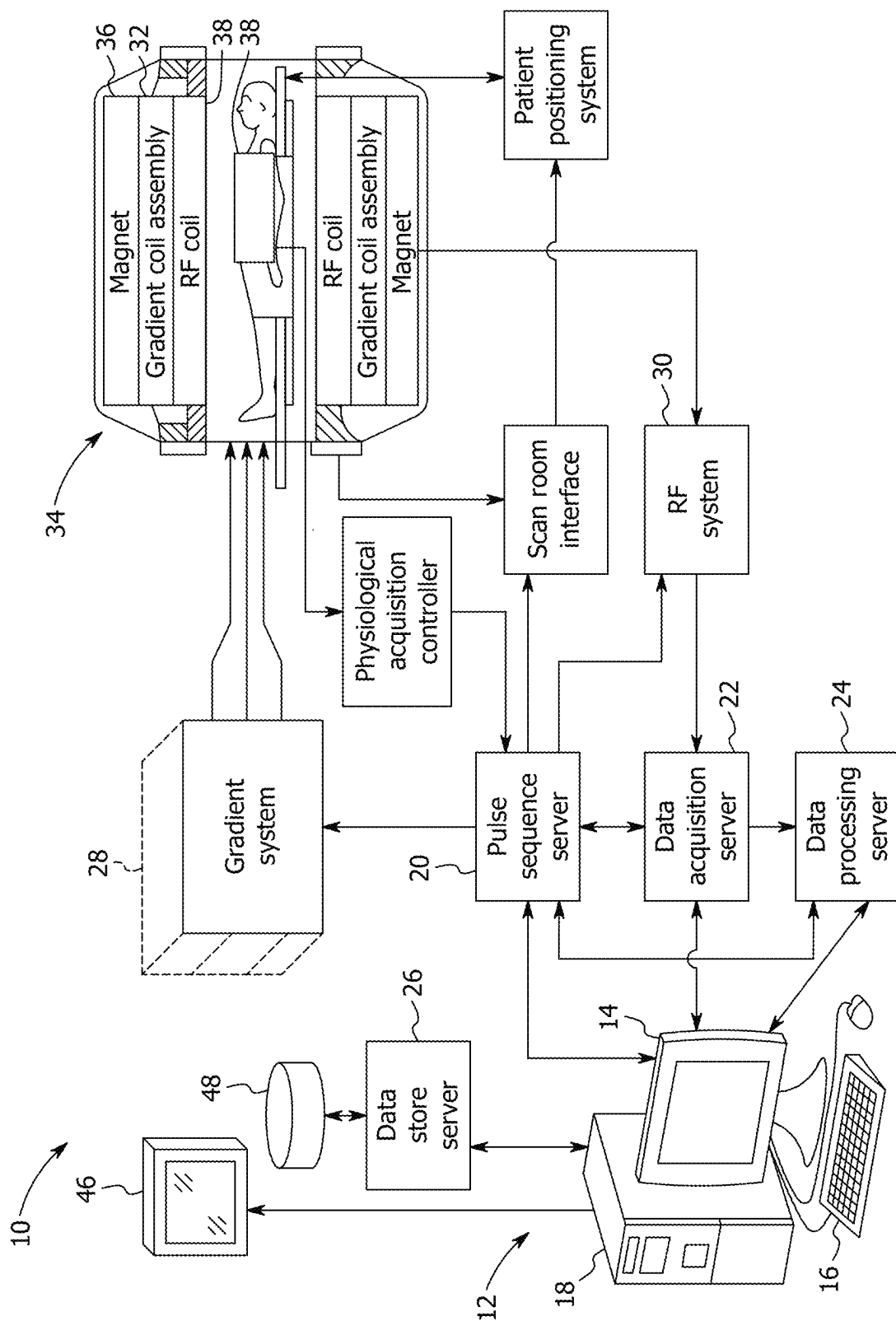


FIG. 1

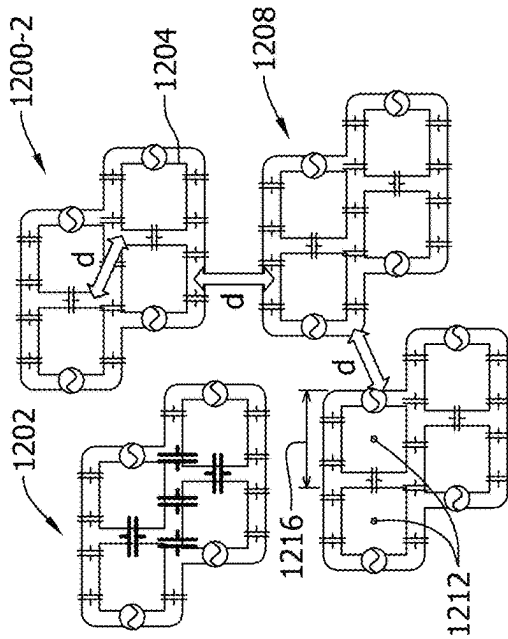


FIG. 2A
PRIOR ART

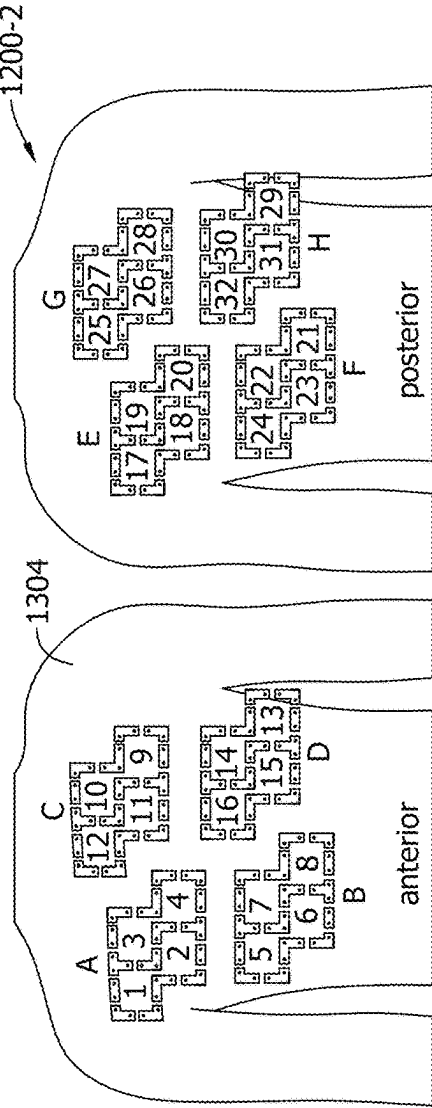


FIG. 2B
PRIOR ART

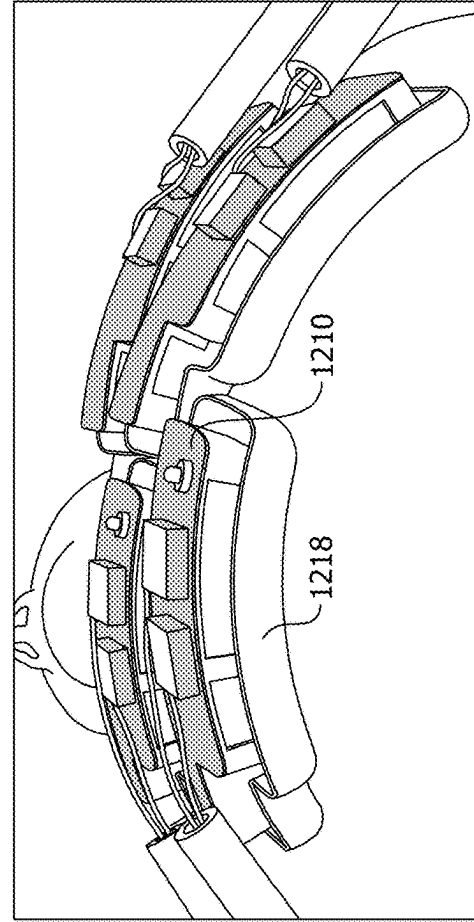


FIG. 2C
PRIOR ART

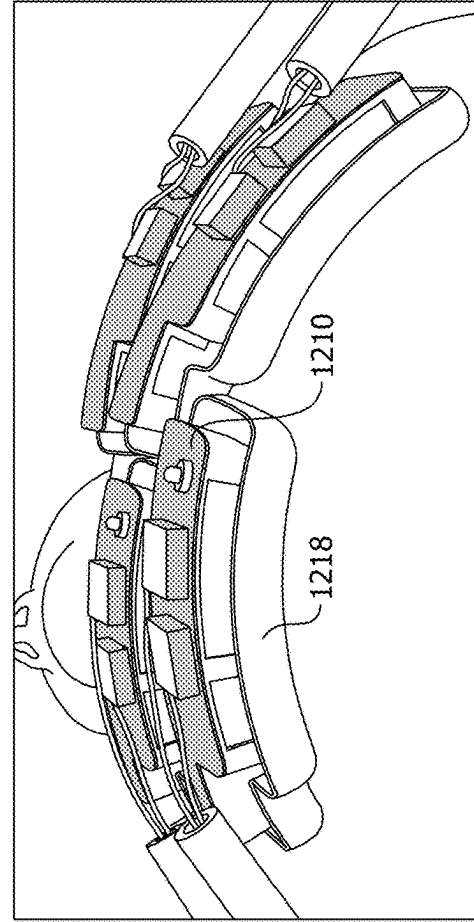


FIG. 2D
PRIOR ART

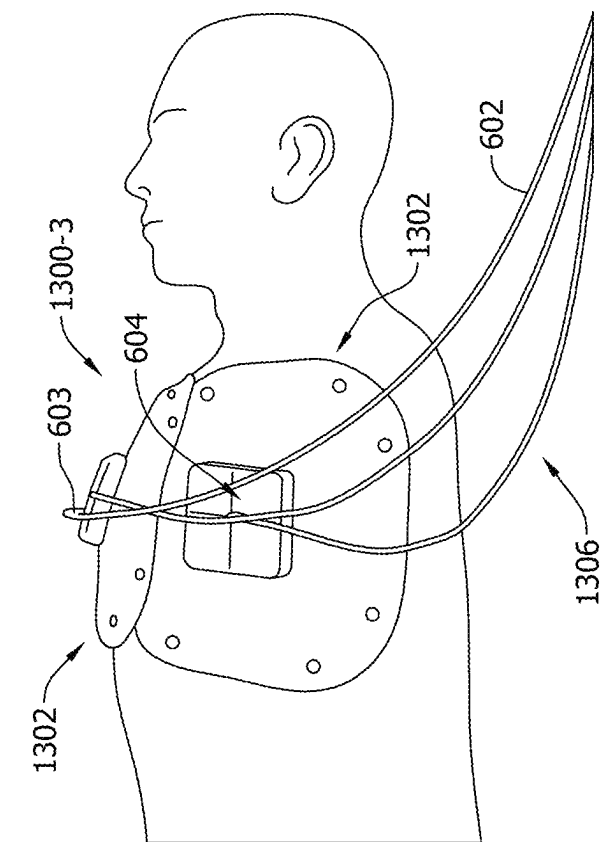


FIG. 3A

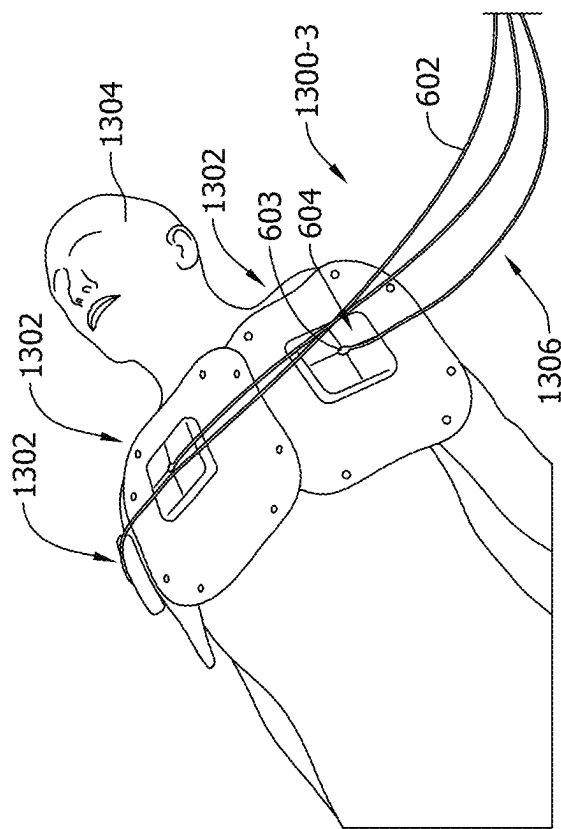


FIG. 3B

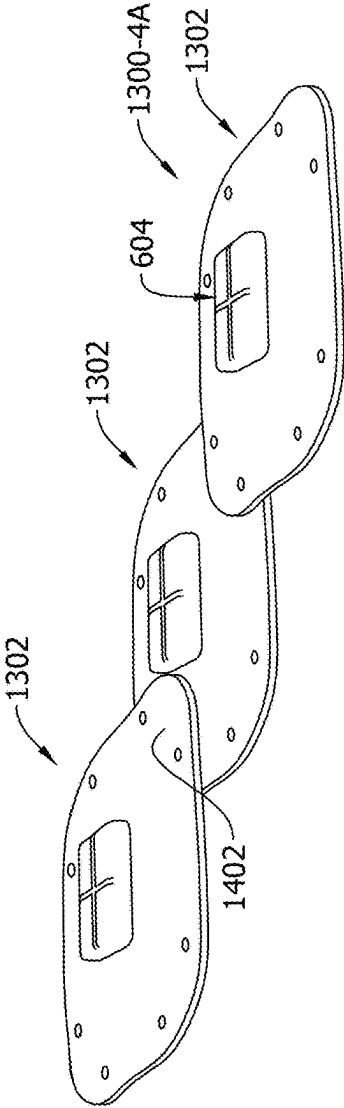


FIG. 4A

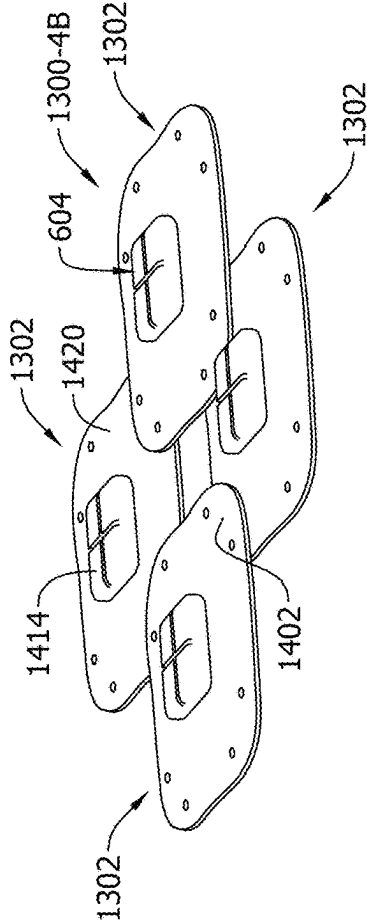


FIG. 4B

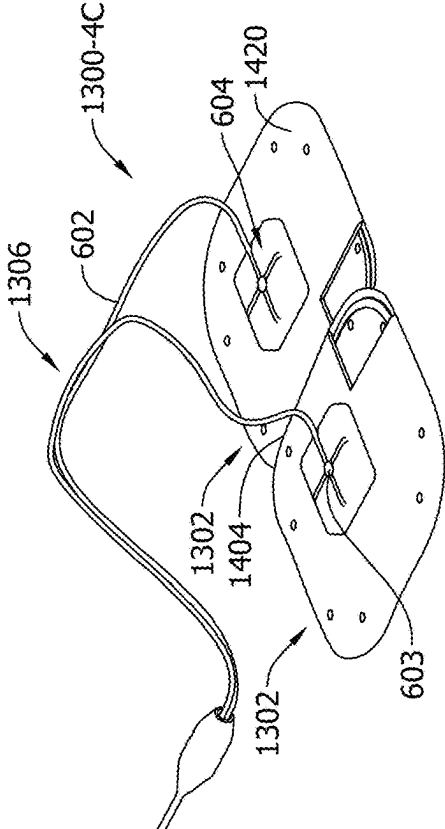


FIG. 4C

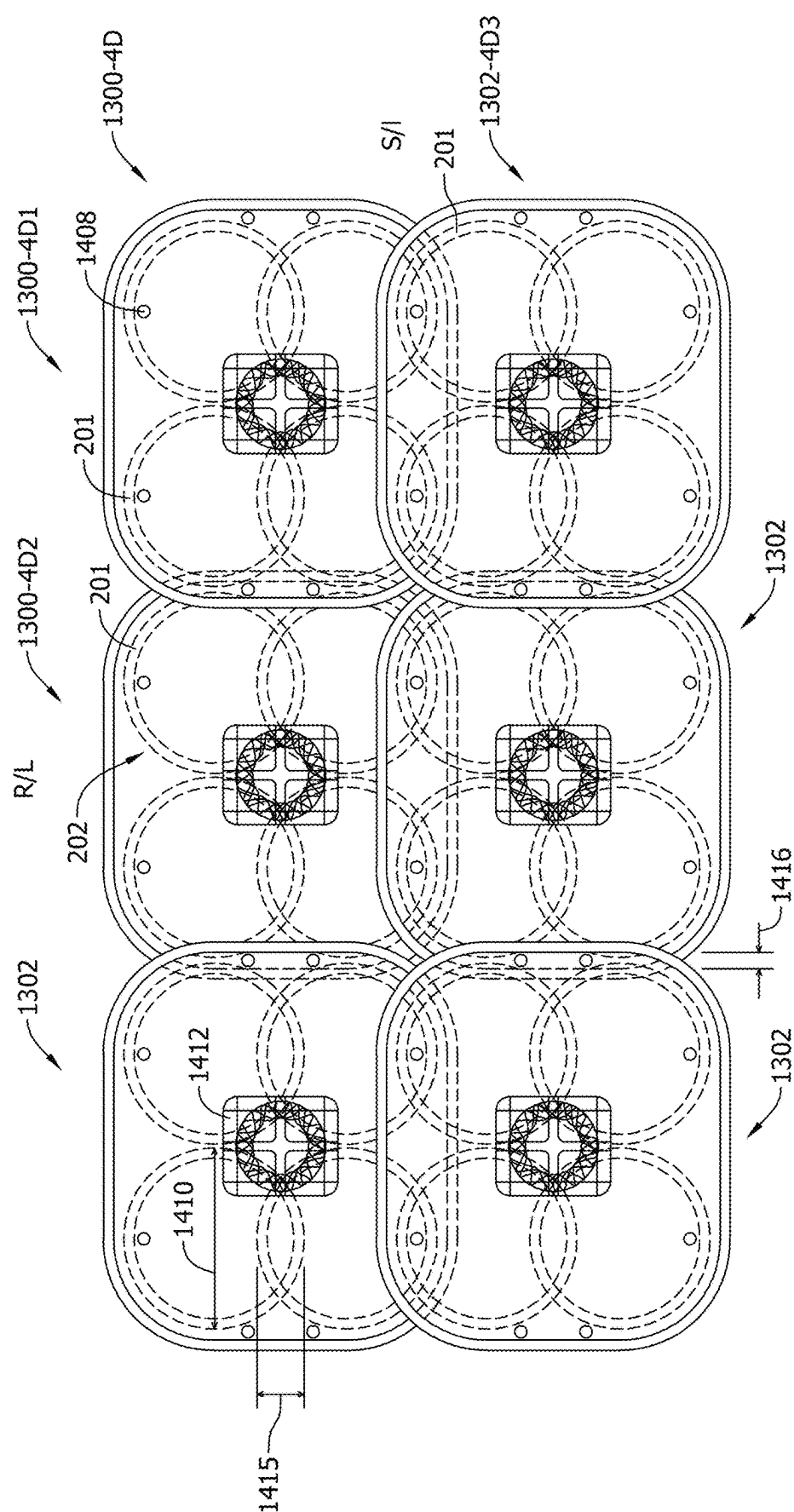


FIG. 4D

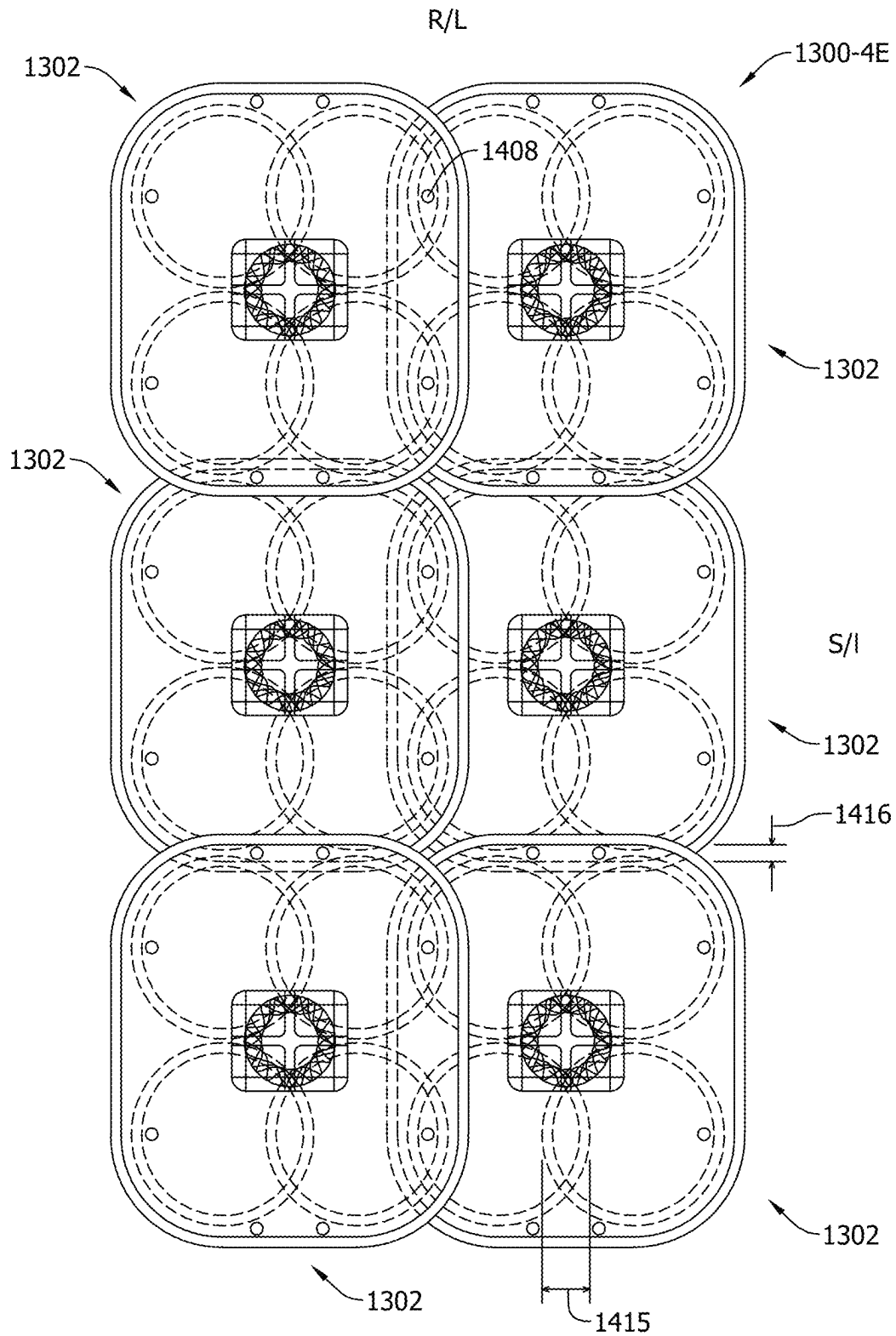


FIG. 4E

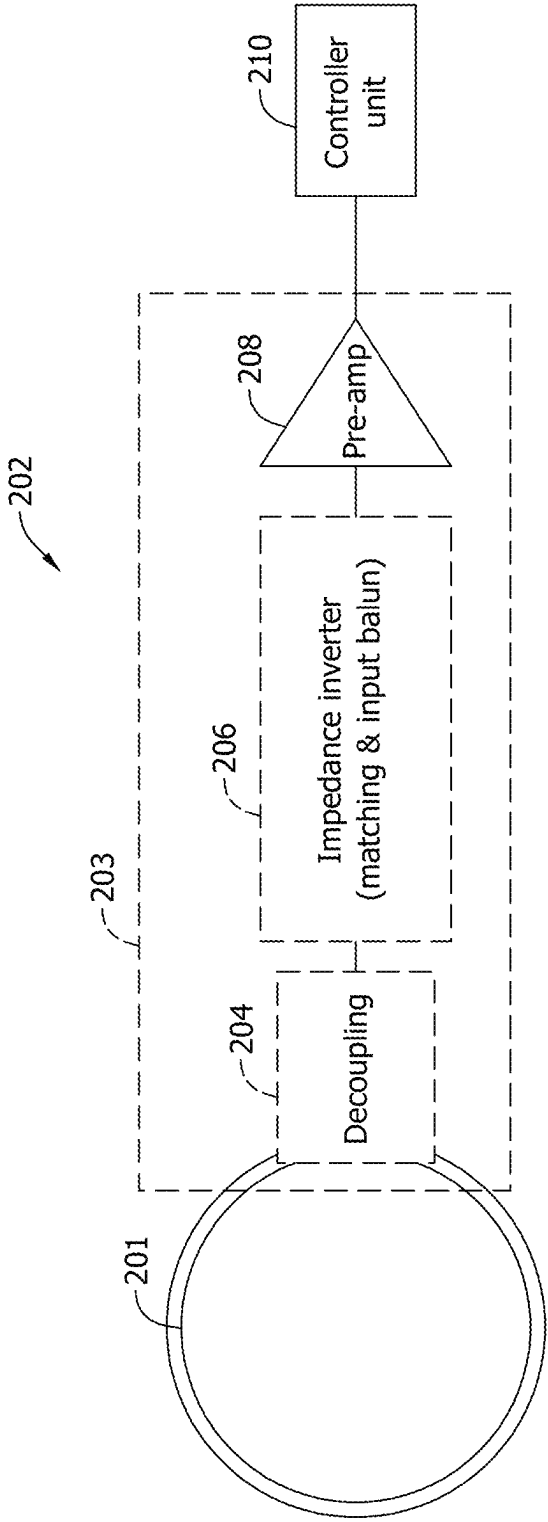


FIG. 5A

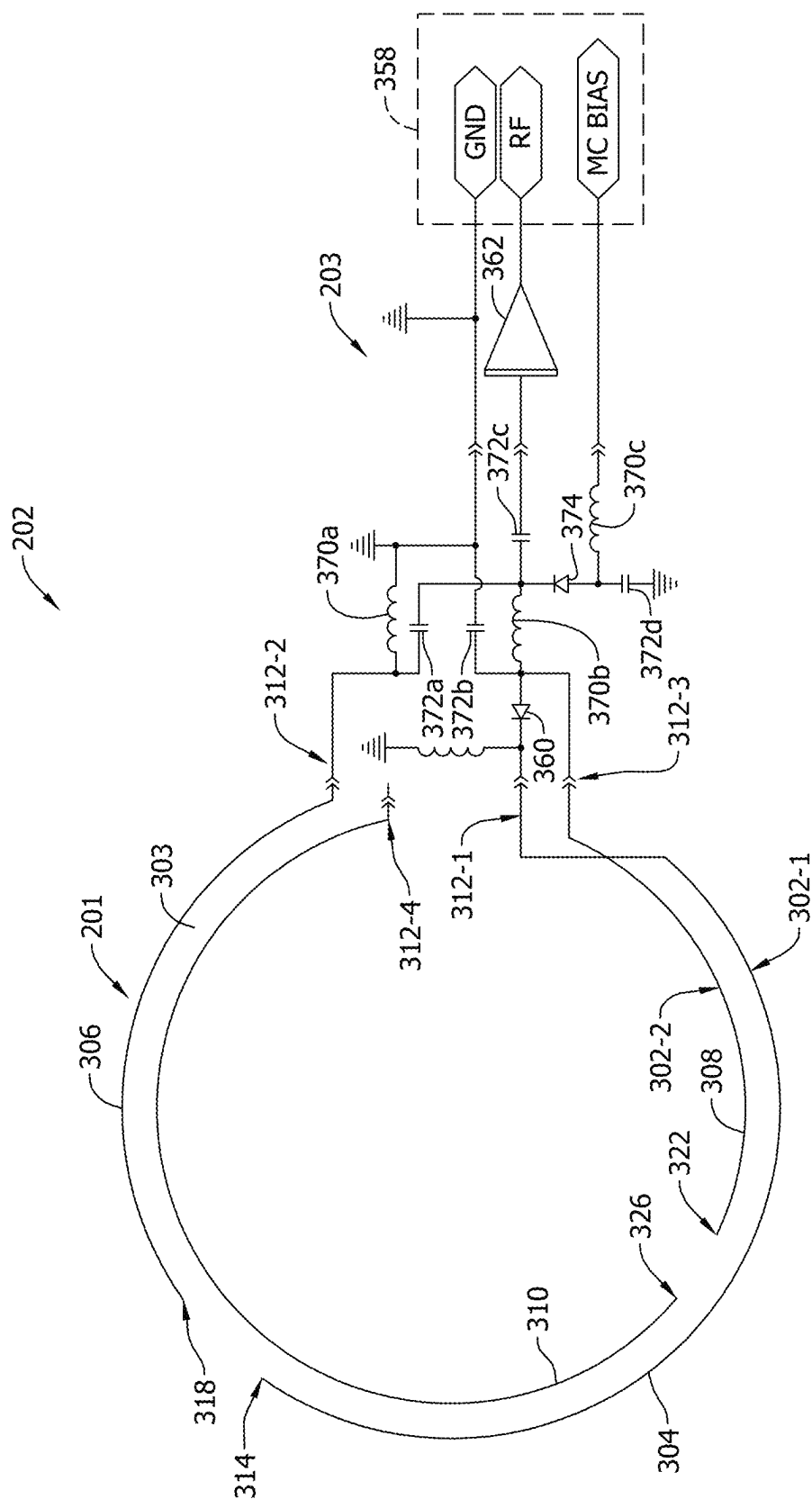


FIG. 5B

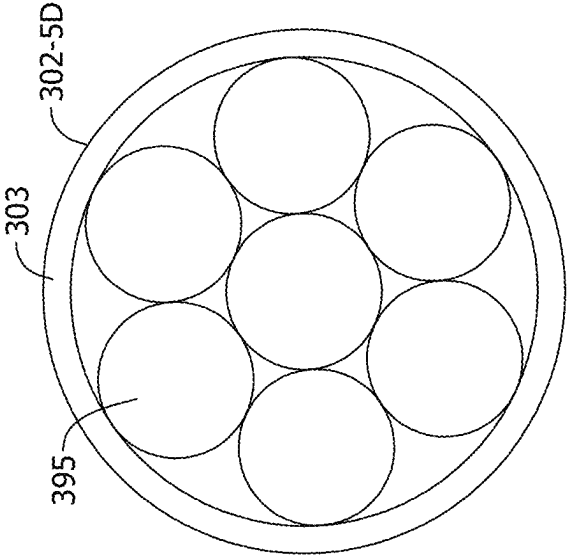


FIG. 5D

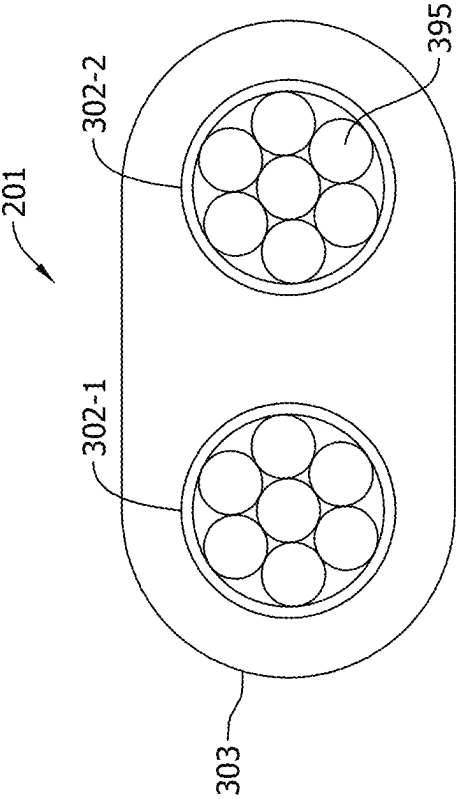


FIG. 5C

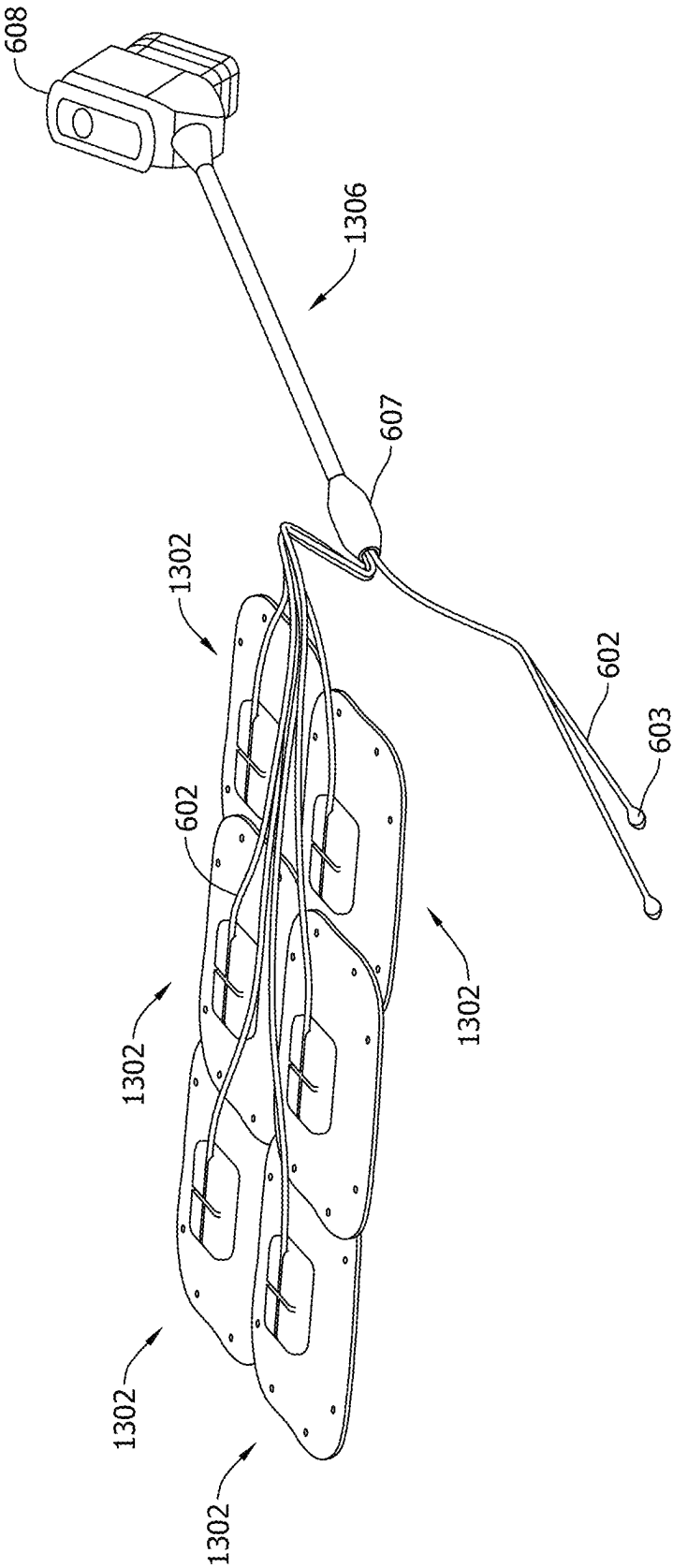


FIG. 6A

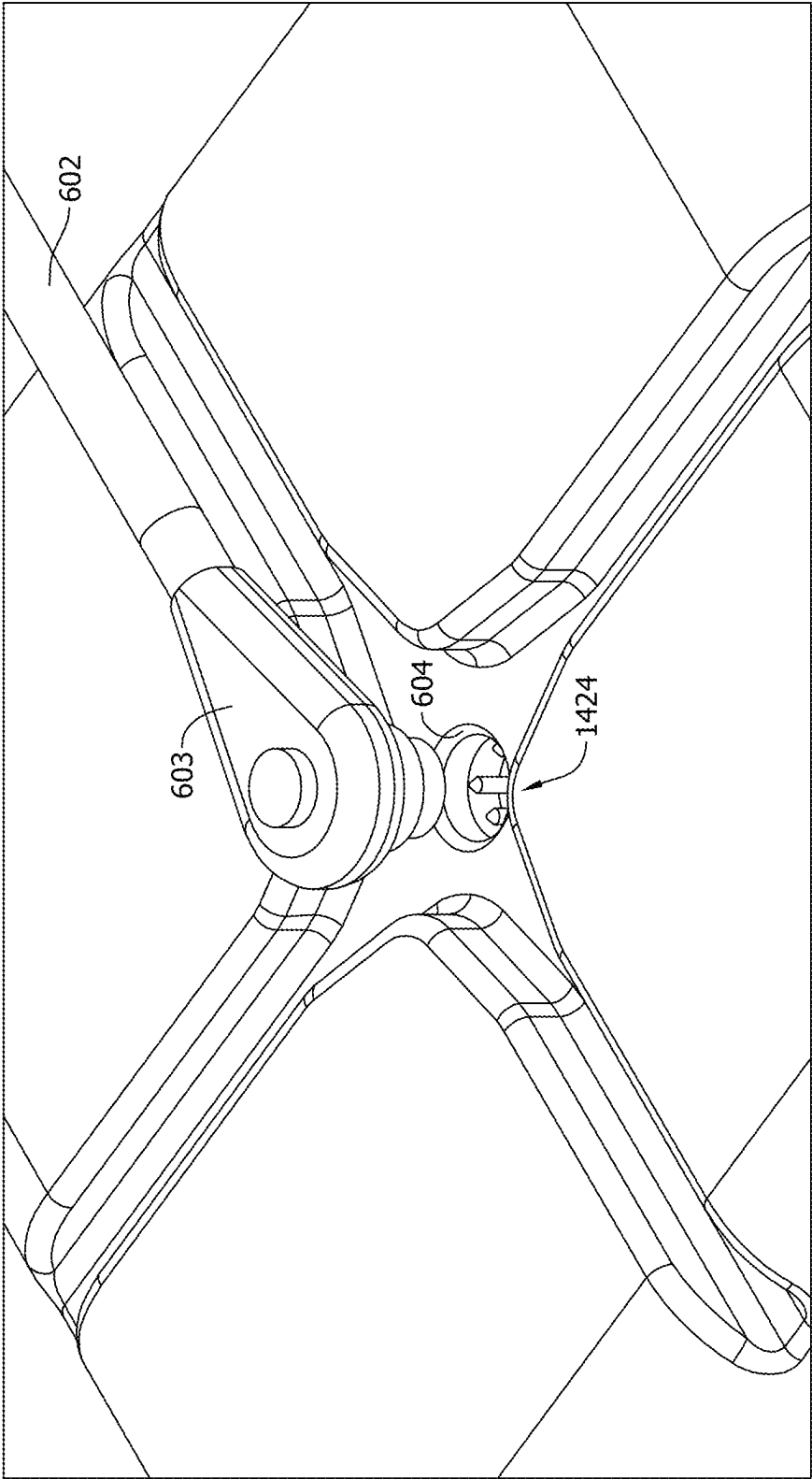


FIG. 6B

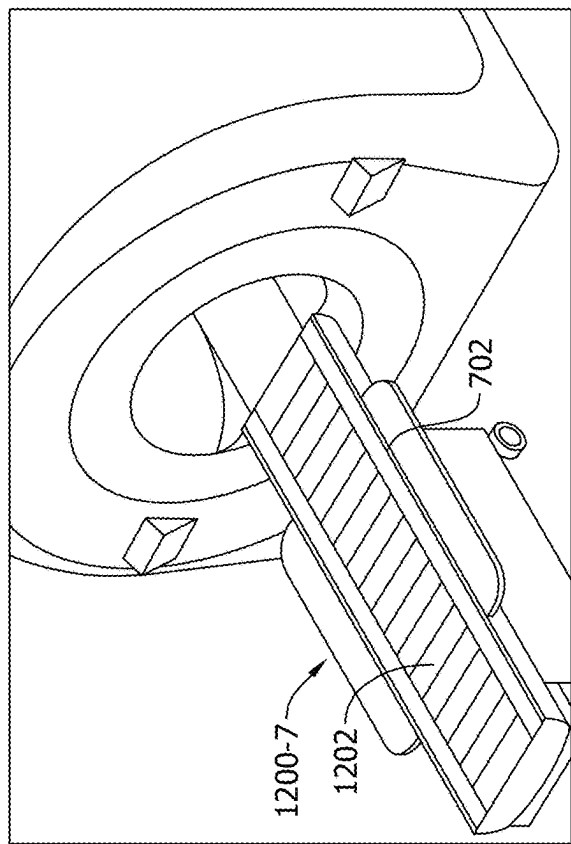


FIG. 7B
PRIOR ART

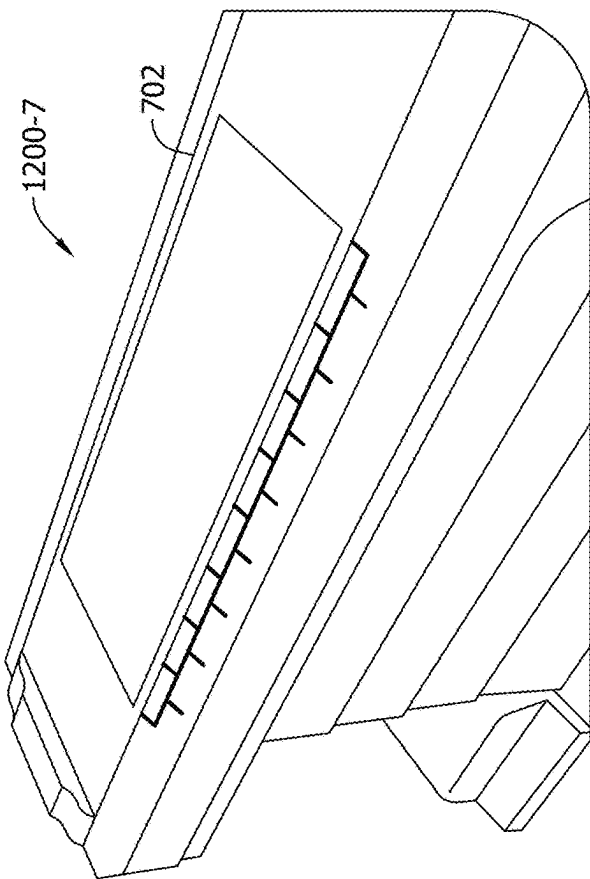


FIG. 7A
PRIOR ART

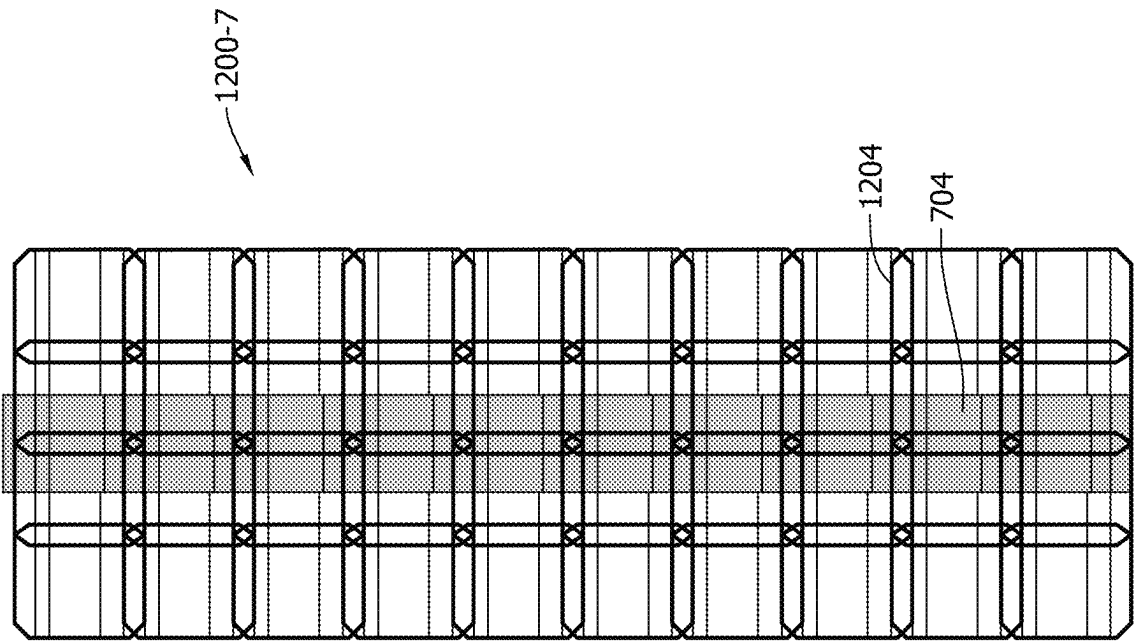


FIG. 7C
PRIOR ART

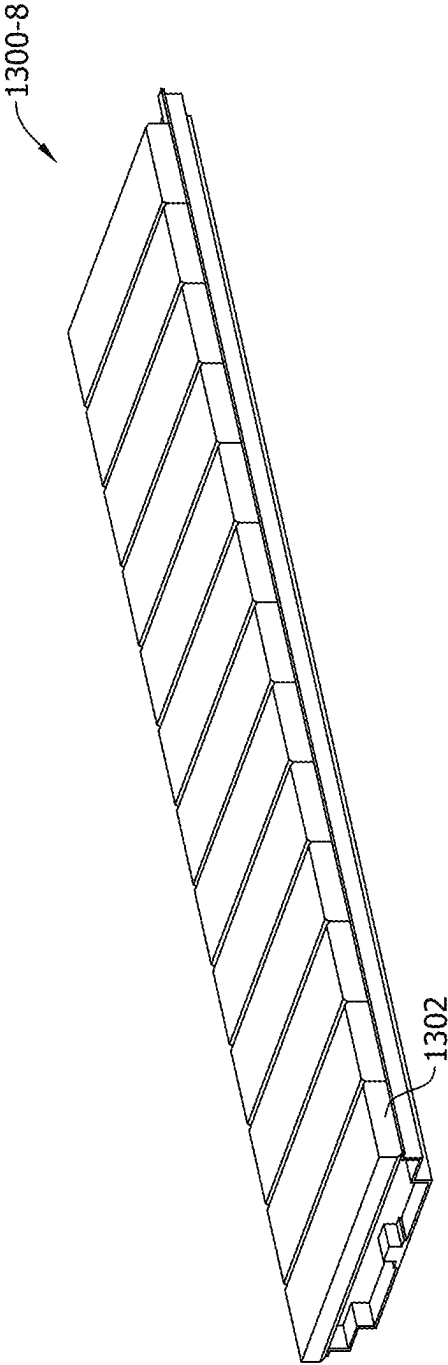


FIG. 8A

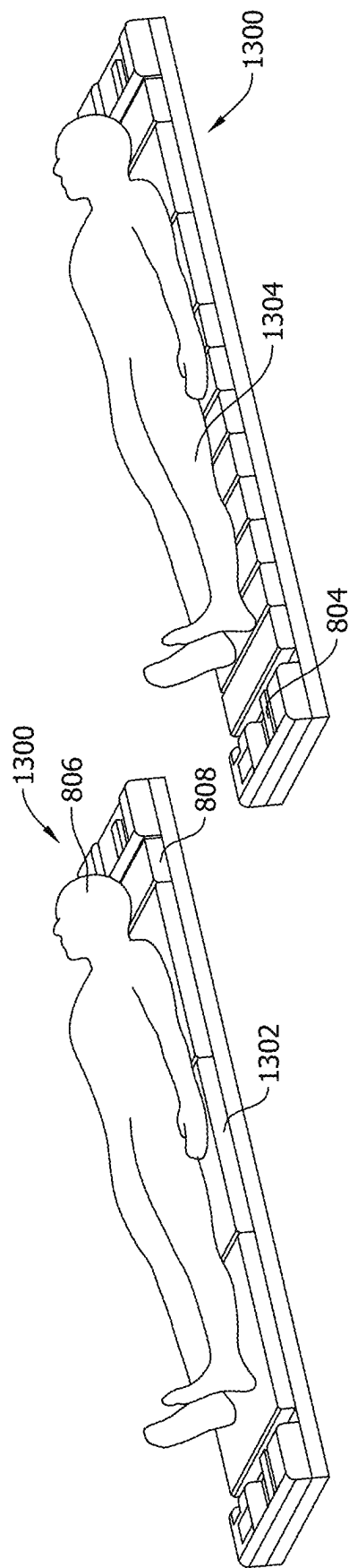


FIG. 8B

FIG. 8C

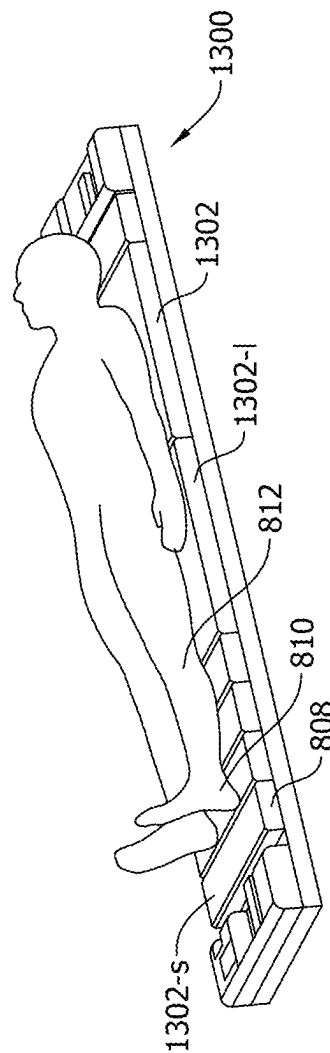


FIG. 8D

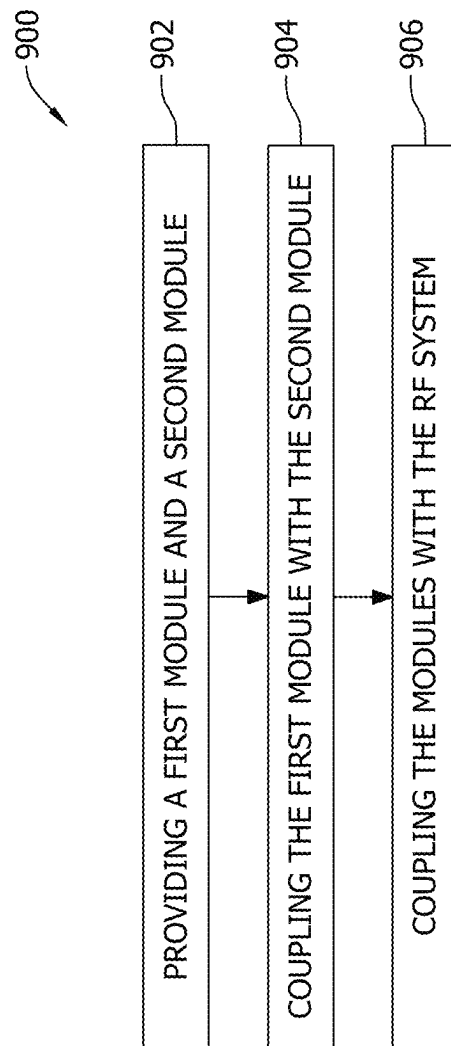


FIG. 9

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MODULAR RADIO-FREQUENCY COIL ASSEMBLIES OF A MAGNETIC RESONANCE SYSTEM

BACKGROUND

The field of the disclosure relates generally to a magnetic resonance (MR) system, and more particularly, to radio frequency (RF) coil assemblies for an MR system.

Magnetic resonance imaging (MRI) has proven useful in diagnosis of many diseases. MRI provides detailed images of soft tissues, abnormal tissues such as tumors, and other structures, which cannot be readily imaged by other imaging modalities, such as computed tomography (CT). Further, MRI operates without exposing patients to ionizing radiation experienced in modalities such as CT and x-rays.

In MRI, an RF coil assembly is used to detect MR signals emitted from a subject and thus is a key component of an MR system. Known RF coil assemblies are disadvantaged in some aspects and improvements are desired.

BRIEF DESCRIPTION

In one aspect, a radio frequency (RF) coil assembly for a magnetic resonance (MR) system is provided. The RF coil assembly includes one or more modules. A module of the one or more modules includes one or more RF coils, each RF coil including a coil loop that includes a wire conductor, the wire conductor formed into the coil loop. Each RF coil also includes a coupling electronics portion coupled with the coil loop. The module further includes an outlet coupled with coupling electronics portions of the one or more RF coils. The RF coil assembly further includes a wiring harness including module interfacing cables each configured to be electrically coupled with the outlet.

In another aspect, an RF coil assembly for an MR system is provided. The RF coil assembly includes a first module and a second module. The first module includes a first number of RF coils, each RF coil including a coil loop. The first module also includes a first outer enclosure housing coil loops of the first number of RF coils. The second module includes a second number of RF coils, the second number being different from the first number. The second module further includes a second outer enclosure housing coil loops of the second number of RF coils.

In one more aspect, a module of an RF coil assembly for an MR system is provided. The module includes one or more RF coils, each RF coil including a coil loop that includes a wire conductor, the wire conductor formed into the coil loop. Each RF coil also includes a coupling electronics portion coupled with the coil loop. The module further includes an outlet coupled with coupling electronics portions of the one or more RF coils.

BRIEF DESCRIPTION OF DRAWINGS

Non-limiting and non-exhaustive embodiments are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various drawings unless otherwise specified.

FIG. 1 is a block diagram of a magnetic resonance (MR) system.

FIG. 2A is a schematic diagram of electronic components of a known radio frequency (RF) coil assembly.

FIG. 2B is a schematic diagram of placement of coils of the known RF coil assembly shown in FIG. 2A.

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FIG. 2C illustrates the known RF coil assembly shown in FIG. 2A being placed on a subject.

FIG. 2D illustrates the known RF coil assembly shown in FIG. 2A being placed on a subject with top covers of the RF coil assembly removed.

FIG. 3A is a perspective view of an example RF coil assembly placed on a subject.

FIG. 3B is a side view of the RF coil assembly shown in FIG. 3A.

FIG. 4A shows another example RF coil assembly.

FIG. 4B shows one more example RF coil assembly.

FIG. 4C shows one more example RF coil assembly.

FIG. 4D shows one more example RF coil assembly.

FIG. 4E shows one more example RF coil assembly.

FIG. 5A is a block diagram of an example RF coil.

FIG. 5B is a schematic diagram of an example embodiment of the RF coil shown in FIG. 5A.

FIG. 5C is a cross-sectional view of an example coil loop of the RF coil shown in FIG. 5A.

FIG. 5D is a cross-sectional view of another example coil loop of the RF coil shown in FIG. 5A.

FIG. 6A is a perspective view of an example RF coil assembly.

FIG. 6B illustrates a portion of the RF coil assembly shown in FIG. 6A with a wiring harness unplugged from a module.

FIG. 7A shows another known modular coil assembly.

FIG. 7B shows the known coil assembly with the top cover removed.

FIG. 7C shows the coil loops of the known coil assembly.

FIG. 8A shows one more example coil assembly.

FIG. 8B shows one more example coil assembly.

FIG. 8C shows one more example coil assembly.

FIG. 8D shows one more example coil assembly.

FIG. 9 is a flow chart of an example method of assembling an RF coil assembly shown in FIGS. 1, 3A-6B, and 8A-8D.

DETAILED DESCRIPTION

The disclosure includes radio frequency (RF) coil assemblies for use in magnetic resonance (MR) systems for scanning a subject. As used herein, a subject is a human, an animal, a phantom, or any object scanned by a medical imaging system. MR imaging is described as an example only. The assemblies, systems, and methods described herein may be used for MR spectroscopy. MR systems are described as examples only. The RF coil assemblies and methods of assembling RF coil assemblies described herein may be used for medical imaging systems other than MR systems, such as positron emission tomography (PET)-MR systems. Method aspects of assembling and using the RF coil assemblies will be in part apparent and in part explicitly discussed in the following description.

In MR imaging (MRI), a subject is placed in a magnet. When the subject is in the magnetic field generated by the magnet, magnetic moments of nuclei, such as protons, attempt to align with the magnetic field but precess about the magnetic field in a random order at the nuclei's Larmor frequency. The magnetic field of the magnet is referred to as B_0 and extends in the longitudinal or z direction. In acquiring an MR image, a magnetic field (referred to as an excitation field B_1), which is in the x-y plane and near the Larmor frequency, is generated by an RF coil and may be used to rotate, or "tip," the net magnetic moment M_z of the nuclei from the z direction to the transverse or x-y plane. A signal, which is referred to as an MR signal, is emitted by the nuclei, after the excitation signal B_1 is terminated. To use

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the MR signals to generate an image of a subject, magnetic field gradient pulses (G_x , G_y , and G_z) are used. The gradient pulses are used to scan through the k-space, the space of spatial frequencies or inverse of distances. A Fourier relationship exists between the acquired MR signals and an image of the subject, and therefore the image of the subject can be derived by reconstructing the MR signals.

FIG. 1 illustrates a schematic diagram of an example MR system 10. In the example embodiment, MR system 10 includes a workstation 12 having a display 14 and a keyboard 16. Workstation 12 includes a processor 18, such as a commercially available programmable machine running a commercially available operating system. Workstation 12 provides an operator interface that allows scan prescriptions to be entered into MR system 10. Workstation 12 is coupled to a pulse sequence server 20, a data acquisition server 22, a data processing server 24, and a data store server 26. Workstation 12 and each server 20, 22, 24, and 26 communicate with each other.

In the example embodiment, pulse sequence server 20 responds to instructions downloaded from workstation 12 to operate a gradient system 28 and an RF system 30. The instructions are used to produce gradient and RF waveforms in MR pulse sequences. An RF coil assembly 38 and a gradient RF coil assembly 32 are used to perform the prescribed MR pulse sequence. RF coil assembly 38 may be a whole body RF coil. RF coil assembly 38 may also be a local RF coil assembly 38 that may be placed in proximity to the anatomy to be imaged, or a coil array that includes a plurality of coils.

In the example embodiment, gradient waveforms used to perform the prescribed scan are produced and applied to gradient system 28, which excites gradient coils in gradient RF coil assembly 32 to produce the magnetic field gradients G_x , G_y , and G_z used for position-encoding MR signals. Gradient RF coil assembly 32 forms part of a magnet assembly 34 that also includes a polarizing magnet 36 and RF coil assembly 38.

In the example embodiment, RF system 30 includes an RF transmitter for producing RF pulses used in MR pulse sequences. The RF transmitter is responsive to the scan prescription and direction from pulse sequence server 20 to produce RF pulses of a desired frequency, phase, and pulse amplitude waveform. The generated RF pulses may be applied to RF coil assembly 38 by RF system 30. Responsive MR signals detected by RF coil assembly 38 are received by RF system 30, amplified, demodulated, filtered, and digitized under direction of commands produced by pulse sequence server 20. RF coil assembly 38 is described as a transmit and receive coil such that RF coil assembly 38 transmits RF pulses and detects MR signals. In one embodiment, MR system 10 may include a transmit RF coil that transmits RF pulses and a separate receive coil that detects MR signals. A transmission channel of RF system 30 may be connected to an RF transmit coil and a receiver channel may be connected to a separate RF receive coil. Often, the transmission channel is connected to the whole body RF coil assembly 38 and each receiver section is connected to a separate local RF coil.

In the example embodiment, RF system 30 also includes one or more RF receiver channels. Each RF receiver channel includes an RF amplifier that amplifies the MR signal received by RF coil assembly 38 to which the channel is connected, and a detector that detects and digitizes the I and Q quadrature components of the received MR signal. The magnitude of the received MR signal may then be deter-

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mined as the square root of the sum of the squares of the I and Q components as in Eq. (1) below:

$$M = \sqrt{I^2 + Q^2}; \quad (1)$$

and the phase of the received MR signal may also be determined as in Eq. (2) below:

$$\varphi = \tan^{-1}\left(\frac{Q}{I}\right) \quad (2)$$

In the example embodiment, the digitized MR signal samples produced by RF system 30 are received by data acquisition server 22. Data acquisition server 22 may operate in response to instructions downloaded from workstation 12 to receive real-time MR data and provide buffer storage such that no data is lost by data overrun. In some scans, data acquisition server 22 does little more than pass the acquired MR data to data processing server 24. In scans that need information derived from acquired MR data to control further performance of the scan, however, data acquisition server 22 is programmed to produce the needed information and convey it to pulse sequence server 20. For example, during prescans, MR data is acquired and used to calibrate the pulse sequence performed by pulse sequence server 20. Also, navigator signals may be acquired during a scan and used to adjust the operating parameters of RF system 30 or gradient system 28, or to control the view order in which k-space is sampled.

In the example embodiment, data processing server 24 receives MR data from data acquisition server 22 and processes it in accordance with instructions downloaded from workstation 12. Such processing may include, for example, Fourier transformation of raw k-space MR data to produce two or three-dimensional images, the application of filters to a reconstructed image, the performance of a back-projection image reconstruction of acquired MR data, the generation of functional MR images, and the calculation of motion or flow images.

In the example embodiment, images reconstructed by data processing server 24 are conveyed back to, and stored at, workstation 12. In some embodiments, real-time images are stored in a database memory cache (not shown in FIG. 1), from which they may be output to operator display 14 or a display 46 that is located near magnet assembly 34 for use by attending physicians. Batch mode images or selected real time images may be stored in a host database on disc storage 48 or on a cloud. When such images have been reconstructed and transferred to storage, data processing server 24 notifies data store server 26. Workstation 12 may be used by an operator to archive the images, produce films, or send the images via a network to other facilities.

During a scan, interfacing cables may be used to transmit signals between RF coil assembly 38 and other aspects of MR system 10 (e.g., RF system 30, data acquisition server 22, and pulse sequence server 20), for example to control the RF coils and/or to receive signals from the RF coils. As described above, the RF coil assembly 38 may be a transmit coil that transmits RF excitation signals, or a receive coil that receives the MR signals emitted by the subject. In an example, the transmit and receive coils are a single mechanical and electrical structure or array of structures, with transmit/receive mode switchable by auxiliary circuitry. In

other examples, a transmit coil and a receive coil may be independent structures that are physically coupled to each other via the RF system 30. For enhanced image quality, however, a receive coil is desirable to be mechanically and electrically isolated from the transmit coil. In such cases, the receive coil in the receive mode is electromagnetically coupled to and resonant with an RF “echo” that is stimulated by the transmit coil. On the other hand, during a transmit mode, the receive coil is electromagnetically decoupled from and therefore not resonant with the transmit coil, during transmission of the RF signal. Such decoupling averts a potential problem of noise produced within the auxiliary circuitry when the receive coil couples to the full power of the RF signal.

A traditional receive coil for MR includes several conductive intervals joined between themselves by capacitors. By adjusting the capacitors’ values, the impedance of the RF coil may be brought to its minimal value, usually characterized by a low resistance. At a resonant frequency, stored magnetic and electric energy alternate periodically. Each conductive interval, due to its length and width, possesses a certain self-capacitance, where electric energy is periodically stored as static electricity. The distribution of this electricity takes place over the entire conductive interval length in the order of 5-15 cm, causing similar range electric dipole field. In proximity of a large dielectric load, self-capacitance of the intervals change, resulting in detuning of the coil. In case of a lossy dielectric, dipole electric field causes Joule dissipation characterized by an increased overall resistance observed by the coil.

Traditional RF coils may include acid etched copper traces or loops on printed circuit boards (PCBs) with lumped electronic components (e.g., capacitors, inductors, baluns, and resistors), matching circuitry, decoupling circuitry, and pre-amplifiers (see FIGS. 2A-2D described later). Such a configuration is typically bulky, heavy, and rigid, and requires relatively strict placement of the coils relative to each other in an array to prevent coupling interactions among coil elements that may degrade image quality. As such, traditional RF coils and RF coil arrays lack flexibility and therefore may not conform to subject anatomy, degrading imaging quality and subject comfort.

FIGS. 2A-2D show a known RF coil assembly 1200 including modules 1202 constructed with traditional RF coils and RF coil arrays. FIG. 2A is a schematic diagram of the circuit elements of RF coil assembly 1200. FIG. 2B shows placement of modules on a subject 1304. FIG. 2C shows RF coil assembly 1200 being placed with subject 1304. FIG. 2D shows a top portion of RF coil assembly 1200 placed on subject 1304 with top covers 1206 of modules 1202 removed. RF coil assembly 1200 includes eight modules 1202, four modules 1202 to be placed over subject 1304, and four modules 1202 to be placed underneath subject 1304. In each module 1202, four coils 1208 are included. An individual coil 1208 includes a copper strip or copper trace 1210, which is rigid and heavy.

Two coil loops can couple magnetically and electrically. One form of coupling is mutual inductance where signals and noise are transferred from one coil loop to another. Coupling of coil loops introduces artifacts and reduces signal to noise ratio (SNR). The mutual inductance may be reduced by overlapping the coil loops or placing the coil loops at a distance. Coil X is overlapped with coil Y when the area defined by a coil loop of coil X overlaps or intersects with the area defined by a coil loop of coil Y. For example, in module A, coil 1 is overlapped with coils 2 and 3 (see FIGS. 2A and 2B). For traditional coil loops, critical over-

lapping is needed to minimize coupling between adjacent coil when using overlapping to reduce coupling. Two coil loops are critically overlapped when the mutual inductance coupling between the two coil loops is zero or approximately zero. For square coil loops 1204, if centers 1212 of square coil loops 1204 are separated by approximately 0.9 of length 1216 of an edge of square coil loop 1204. Coil loops 1204 are overlapped with one another to reduce mutual inductance coupling. Modules 1202 are underlapped from one another. A coil is underlapped from another coil when the coil is not overlapped with the other coil at any part of the coil. Module X is underlapped with module Y when any coil of module X is underlapped with any coil of module Y. For example, module A is underlapped with modules B, C, or D (see FIGS. 2A and 2B). A minimum distance is required to minimize the coupling between two loops. The minimum distance depends on the coil construction and the tolerance level for coupling. However, when the two coil loops are too far apart, signals of anatomies at the gap between the two coil loops are not received in either coil loop, causing a gap in the image.

Because of the rigid requirement on coil distance and construction, modules 1202 in known RF coil assembly 1200 need to be fitted in a predefined manner like puzzle pieces. If modules 1202 are not assembled in the predefined manner, image quality is compromised. Therefore, nonfunctioning of one module 1202 renders the entire RF coil assembly 1200 nonfunctioning, resulting in decommission of the entire RF coil assembly 1200.

Further, because the rigid requirement on coil distance and construction, rigid structure is needed to keep the formation of the coil. As shown in FIG. 2D, coil loops 1204 are fabricated from copper strips 1210. Copper strips 1210 are fastened to an enclosure 1218 of module 1202. Enclosures 1218 are also rigid to keep the placement and shape of copper strips 1210.

In contrast, the RF coil used in the RF coil assemblies described herein includes a coil loop formed by wire conductors. In the case of two RF coils overlapping, the coupling electronics portion that couples with the coil loop of the RF coil has high blocking or source impedance, thereby minimizing mutual inductance coupling. A thin cross-section of the wire conductor in the RF coil reduces the parasitic capacitance at the cross-overs or overlaps, and reduces other coupling such as electric field coupling and eddy current, in comparison to two traditional trace-based loops. The combination of high blocking impedance and a thin cross-section of the RF coil loop allows flexible placement of multiple coils into one RF coil assembly over a finite area, while coupling between the RF coils is minimized and critical overlapping between two loops is not required. Wire conductors also add flexibility to the coils, allowing the RF coil assembly to conform with a curved anatomy of a subject.

FIGS. 3A and 3B show an example RF coil assembly 1300-3 including modules 1302 that are pluggable and serviceable. FIG. 3A is a perspective view of RF coil assembly 1300 placed on subject 1304. FIG. 3B is a side view of RF coil assembly 1300 placed on subject 1304. In the example embodiment, RF coil assembly 1300 may further include wiring harness 1306. Modules 1302 are coupled with wiring harness 1306. The other end of wiring harness 1306 is coupled with one of system ports of MR system 10. The number of ports in MR system 10 is limited. MR system 10 typically includes four ports, each providing a certain number of channels such as 16, 32, or 64 channels,

or other numbers of channels. Signals from and to RF coils **202** are transmitted via ports to RF system **30**.

FIGS. 4A-4E show more example RF coil assemblies **1300-4A**, **1302-4B**, **1302-4C**, **1302-4D**, and **1302-4E**. In the example embodiments, modules **1302** of RF coil assembly **1300** are coupled with one another at arbitrary orientations. A module **1302** of RF coil assembly **1300-4A**, **1302-4B** is coupled with another module **1302** at corners **1402** of modules **1302**. A module **1302** of RF coil assembly **1300-4C** is coupled with another module along edges **1404** of modules **1302**. Module **1302** is shown as in a square or rectangular shape. Modules **1302** may in other shapes such as circular and polygonal or in irregular shapes. Modules **1302** may be in the same shapes or sizes such that spare modules **1302** may be provided in a scanner room and used to replace a nonfunctioning module **1302** of RF coil assembly **1300**. Modules **1302** may be in different shapes or sizes and spare modules **1302** of corresponding shapes or sizes may be provided for replacement.

In the example embodiments, modules **1302** are not restricted at a predefined distance with one another. A module **1302** may be overlapped with another module **1302** in RF coil assembly **1300-3**, **1302-4A**, **1302-4B**, and **1302-4C**. Module X is overlapped with module Y when a coil of module X is overlapped with a coil of module Y. RF coil assembly **1300** may further include mechanical features such as couplers **1408** (see FIG. 4D) on modules **1302** that allow modules **1302** to be connected to one another to maintain the overlap distance between modules **1302** or keep modules **1302** in place during the use of RF coil assembly **1300**.

In the example embodiments, RF coil assembly **1300-4D**, **1302-4E** may include modules **1302** that are underlapped. Underlapped modules **1302** may be in the S/I (superior/inferior) direction (see FIG. 4D) or in the R/L (right/left) direction (see FIG. 4E). For example, module **1302-4D1** is underlapped with module **1302-4D2**. Coil loops **201** in module **1302-4D1** are underlapped with coil loops **201** in module **1302-4D2**. The gap **1416** between neighboring coil loops **201** of underlapped modules **1302** may be 1 cm with a diameter **1410** of coil loop **201** as 13.6 cm. Modules **1302** of RF coil assembly **1300-4D**, **1302-4E** may be overlapped in the other direction. For example, modules **1302** in the S/I direction are underlapped, while modules **1302** are overlapped in the R/L direction in RF coil assembly **1300-4D**. In RF coil assembly **1300-4E**, modules **1302** are underlapped in the R/L direction while modules **1302** are overlapped in the S/I direction. In some embodiments, all modules **1302** in RF coil assembly are underlapped.

In the example embodiments, module **1302** may include coupler **1408** (see FIG. 4D). Coupler **1408** is configured to couple modules **1302** together. For example, coupler **1408** may be a hook-and-loop fastener. Coupler **1408** also serves as a marker indicating positions at which module **1302** critically overlaps with another module **1302**. When a coupler **1408** of module **1302** is aligned with a coupler **1408** of another module **1302**, the two modules **1302** are critically overlapped, where the coils along the borders of modules **1302** are critically overlapped with one another. For circular coil loops, when one RF coil loop **201** overlaps with another RF coil loop **201** by approximately 25%, where the overlapped distance **1415** between the two RF coil loops is approximately 25% of the diameter **1410** of the RF coil loop **201**, the two RF coil loops **201** are critically overlapped, where the mutual inductance coupling is zero or approximately zero. For circular coil loops **201** having diameters **1410** of 136 cm, an overlapped distance **1415** of 36 cm of

coil loops **201** provides coil loops **201** with critical overlapping, where the coupling is minimized. Other orientations provide critical overlapping because coil loops are circular.

In the example embodiments, module **1302** includes one or more RF coils **202** (see FIG. 4D and FIGS. 5A-5D described later). RF coil **202** includes a coil loop **201** and a coupling electronics portion **203**. Coupling electronics portion **203** or part of coupling electronics portion **203** of RF coils **202** in a module **1302** may be combined into one coupling electronics unit **1412** (see FIGS. 4D and 4E). Coupling electronics unit **1412** includes an electronics box **1414** housing coupling electronics portions **203** or part of coupling electronics portions **203** (see FIG. 4B). The electronic components of coupling electronics unit **1412** may be formed as one PCB. Alternatively, the electronic components may be formed on multiple PCBs. Coupling electronics unit **1412** further includes an outlet **604** sized to be coupled with wiring harness **1306** such that an module interfacing cable **602** of wiring harness is pluggable with outlet **604** (see FIGS. 3A-4C and FIGS. 6A and 6B described later). Outlet **604** may be a male or female outlet, and complementary to module interfacing cable **602** of wiring harness **1306**. As a result, module **1302** is coupled with wiring harness **1306** by plugging module interfacing cable **602** with outlet **604** of module **1302**, greatly increasing assembling convenience.

In the example embodiments, module **1302** includes an outer enclosure **1420** (see FIGS. 4B and 4C). Outer enclosure **1420** may be fabricated from a polyurethane fabric such as DARTEX®. The material for outer enclosure **1420** may be waterproof, semi-vapor permeable, and anti-fungal treated. The material allows for ease of cleaning, and protects the internal electronics from getting wet or soiled. Further, the material is biocompatible and does not irritate the skin of a human subject, and therefore suitable for medical uses. The material is also lighted weighted and flexible. Underneath outer enclosure **1420**, module **1302** may include an inner enclosure (not shown). The inner enclosure covers RF coils **202**. The inner enclosure may be fabricated from a material that provides padding, spacing, and/or flame-retardant properties, such as NOMEX®. Outer enclosure **1420** may further include fasteners, for example hook-and-loop fasters or adhesive patches, to hold modules in place after modules are coupled with one another.

In the example embodiments, coupling electronics unit **1412** is positioned exterior of outer enclosure **1420**. Alternatively, coupling electronics unit **1412** is positioned inside outer enclosure **1420**, with outlet **604** of coupling electronics unit **1412** extending through outer enclosure **1420** and being exposed to provide connection with wiring harness **1306**.

In operation, to assembly RF coil assembly, modules **1302** are placed adjacent to one another at a desired orientation to cover a desired area of subject **1304**. Module interfacing cables **602** of wiring harness **1306** are coupled with outlets **604** of modules **1302**. Coil assembly interfacing connector **608** of wiring harness **1306** is plugged in a port of MR system.

In some embodiments, module **1302** includes a wireless communication unit **1424** configured to communicate with RF system **30** and/or other components of MR system **10** in a wireless mode (see FIG. 6B described later). Module **1302** wirelessly transmits and receives signals to and from RF system **30** and/or other components. Module **1302** may be powered by a battery assembly.

FIGS. 5A-5D show example RF coils **202** of module **1302**. FIG. 5A is a block diagram of RF coil **202**. FIG. 5B is a schematic diagram of RF coil **202**. FIG. 5C is a

cross-sectional view of example coil loop **201** of RF coil **202**. FIG. 5D is a cross-sectional view of another example coil loop **201** of RF coil **202**.

In the example embodiment, RF coil **202** includes coil loop **201** coupled to a controller unit **210** via a coupling electronics portion **203** and a coil-interfacing cable **212**. In one example, the RF coil may be a surface receive coil, which may be single- or multi-channelled. RF coil **202** may operate at one or more frequencies in MR system **10**. Coil-interfacing cable **212** may be a coil-interfacing cable extending between coupling electronics portion **203** or an module interfacing cable **602** extending between module **1302** and other components of MR system **10** such as RF system **30** (see FIGS. 6A and 6B described later).

As described above, one form of coupling is mutual inductance where signals and noise are transferred from one coil loop to another. The mutual inductance may be reduced by overlapping the coil loops. The mutual inductance may also be reduced by using a high blocking impedance in the coupling electronics portion. The blocking impedance R_{block} seen by the coil loop in general depends on the resistance R of the coil loop, matching characteristic impedance Z_0 of the transmission line, and input impedance of the LNA (a linear amplifier or preamplifier) R_{ina} and may be approximated as:

$$R_{block} = \frac{Z_0 R}{R_{ina}}.$$

When a relatively-high blocking impedance R_{block} is used,

$$\frac{X_L}{R} \leq \frac{Z_0}{R_{ina}}$$

and the induced current from one coil loop to another is minimized, where $X_L = \omega_0 L$ is the reactance of the coil loop at the resonance frequency of the coil loop.

Other coupling such as coupling through electric field and eddy current may be minimized by reducing the cross-section of the wire conductor in coil loop **201**.

In the example embodiment, coupling electronics portion **203** may be coupled to coil loop **201** of RF coil **202**. Coupling electronics portion **203** may include a decoupling circuit **204**, impedance inverter circuit **206**, and a pre-amplifier **208**. Decoupling circuit **204** may effectively decouple the RF coil during a transmit operation. Typically, RF coil **202** in the receive mode may be positioned adjacent a body of a subject being imaged by MR system **10** in order to receive echoes of the RF signal transmitted during the transmit mode. If RF coil **202** is not used for transmission, RF coil **202** is decoupled from the RF transmit coil such as the RF body coil when the RF transmit coil is transmitting the RF signal. The decoupling of the receive coil from the transmit coil may be achieved using resonance circuits and PIN diodes, microelectromechanical systems (MEMS) switches, or another type of switching circuitry. The switching circuitry may activate detuning circuits operatively connected to RF coil **202**.

In the example embodiment, the impedance inverter circuit **206** may form an impedance matching network between RF coil **202** and pre-amplifier **208**. Impedance inverter circuit **206** is configured to transform a coil impedance of RF coil **202** into an optimal source of impedance for pre-amplifier **208**. The impedance inverter circuit **206** may

include an impedance matching network and an input balun. Pre-amplifier **208** receives MR signals from corresponding RF coil **202** and amplifies the received MR signals. In one example, the pre-amplifier may have a low input impedance that is configured to accommodate a relatively high blocking or source impedance. Additional details regarding the RF coil and associated coupling electronics portion will be explained in more detail below with respect to FIG. 5B. Coupling electronics portion **203** may be packaged in a small PCB with a surface area of approximately 2 cm² or smaller. The PCB may be protected with a conformal coating or an encapsulating resin.

Coil-interfacing cable **212**, such as an module interfacing cable, may be used to transmit signals between the RF coils and other components of MR system **10**. The module interfacing cables may be disposed within the bore or imaging space of MR system **10** and subjected to electromagnetic fields produced and used by MR system **10**. In MR systems, coil-interfacing cables **212** may support transmitter-driven common-mode currents, which may in turn create field distortions and/or unpredictable heating of components. Typically, common-mode currents are blocked by using baluns. Baluns or common-mode traps provide high common-mode impedances, which in turn reduces the effect of transmitter-driven currents.

In the example embodiment, coil-interfacing cable **212** may include one or more baluns. In traditional coil-interfacing cables, baluns are positioned with a relatively high density, because high dissipation/voltages may develop if the balun density is too low or if baluns are positioned at an inappropriate location. However, the dense placement may adversely affect flexibility, cost, and performance. As such, the one or more baluns in the coil-interfacing cable may be continuous baluns to ensure no high currents or standing waves, independent of positioning. The continuous baluns may be distributed, flutter, and/or butterfly baluns.

FIG. 5B is a schematic diagram of an example RF coil **202** having segmented conductors formed in accordance with an embodiment. RF coil **202** is a non-limiting example of RF coil **202** shown in FIG. 5A and as such includes coil loop **201** and coupling electronics portion **203**. In the example embodiment, the coupling electronics portion allows the RF coil to transmit and/or receive RF signals when driven by RF system **30** (shown in FIG. 1). In the illustrated embodiment, RF coil **202** includes a first conductor **302-1** and a second conductor **302-2**. First and second conductors **302-1**, **302-2** may be segmented such that the conductors form an open circuit (e.g., form a monopole). The segments of conductors **302-1**, **302-2** may have different lengths. The length of first and second conductors **302-1**, **302-2** may be varied to achieve a select distributed capacitance (DCAP), and accordingly, a select resonance frequency. DCAP, as used herein, represents a capacitance exhibited between conductors that grows evenly and uniformly along the length of the conductors and is void of discrete or lumped capacitive components and discrete or lumped inductive components.

In the example embodiment, first conductor **302-1** includes a first segment **304** and a second segment **306**. First segment **304** includes a driven end **312-1** at an interface terminating to coupling electronics portion **203**. First segment **304** also includes a floating end **314** that is detached from a reference ground, thereby maintaining a floating state. Second segment **306** includes a driven end **312-2** at the interface terminating to the coupling electronics portion and a floating end **318** that is detached from a reference ground.

In the example embodiment, second conductor **302** includes a first segment **308** and a second segment **310**. First

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segment **308** includes a driven end **312-3** at the interface. First segment **308** also includes a floating end **322** that is detached from a reference ground, thereby maintaining a floating state. Second segment **310** includes a driven end **312-4** at the interface, and a floating end **326** that is detached from a reference ground. Driven end **312-4** may terminate at the interface such that end **312-4** is only coupled to the first conductor through the distributed capacitance. The capacitors shown around the loop between the conductors represent the capacitance between the wire conductors.

In the examples herein, the capacitance may grow in a uniform manner along the length of first and second conductors **302-1**, **302-2**. For example, first conductor **302-1** exhibits a distributed capacitance that grows based on the length of first and second segments **304**, **306**. Second conductor **302-2** exhibits a distributed capacitance that grows based on the length of first and second segments **308**, **310**. First segments **304**, **308** may have a different length than second segments **306**, **310**. The relative difference in lengths between first segments **304**, **308** and second segments **306**, **310** may be used to produce an effective LC circuit having a resonance frequency at the desired center frequency. For example, by varying the length of first segments **304**, **308** relative to the lengths of second segments **306**, **310**, an integrated distributed capacitance may be varied.

In the illustrated embodiment, the first and second wire conductors **302-1**, **302-2** are shaped into a coil loop that terminates to an interface. The coil loop defines a conductive pathway along the first and second conductors. The first and second conductors are void of any discrete or lumped capacitive or inductive elements along an entire length of the conductive pathway. FIG. 5B shows first and second conductors **302-1**, **302-2** are segmented as an example. The coil loop may include loops of varying gauge of stranded or solid conductor wire, loops of varying diameters with varying lengths of first and second conductors **302-1**, **302-2**, and/or loops of varying spacing between first and second conductors **302-1**, **302-2**. For example, each of the first and second conductors may have no cuts or gaps (no segmented conductors, not shown) or one or more cuts or gaps (segmented conductors, FIG. 5B) at various locations along the conductive pathway.

In the example embodiment, circular coil loop **201** is depicted as an example only. Coil loop **201** may in other shapes, such as oval, irregularly curved, or rectangular, that enable coil loop **201** to function as described herein. In one example, coil loop **201** is fabricated from a flexible 1.3 millimeter (mm) diameter conductor optimized for zero reactance at 127.73 MHz, the resonance frequency of a 3 T MR system. RF coils **202** may be designed for an MR system **10** having a different field strength, such as 1.5 T. Because wire conductor **302** of coil loop **201** is flexible, the shape of coil loop **201** may change and be deformed to conform to a curved anatomy of the subject, such as deforming from being circular to other shapes such as oval, elliptical, or irregular shapes like Pringles® chips.

In the example embodiment, a dielectric material **303** encapsulates and separates first and second conductors **302-1**, **302-2**. Dielectric material **303** may be selectively chosen to achieve a select distributive capacitance. Dielectric material **303** may be based on a desired permittivity ϵ to vary the effective capacitance of the coil loop. For example, the dielectric material **303** may be air, rubber, plastic, or any other dielectric material. In one example, the dielectric material may be polytetrafluoroethylene (PTFE). Dielectric material **303** may be an insulating material surrounding the

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parallel conductive elements of first and second conductors **302-1**, **302-2**. Alternatively, first and second conductors **302-1**, **302-2** may be twisted upon one another to form a twisted pair cable. As another example, dielectric material **303** may be a plastic material. First and second conductors **302-1**, **302-2** may form a coaxial structure in which dielectric material **303** separates the first and second conductors. As one more example, the first and second conductors may be configured as planar strips.

In the example embodiment, the coupling electronics portion **203** is operably and communicatively coupled to RF system **30** to allow RF coil **202** to transmit and/or receive RF signals. In the illustrated embodiment, coupling electronics portion **203** includes a signal interface **358** configured to transmit and receive the RF signals. Signal interface **358** may transmit and receive the RF signals via a cable. The cable may be a 3-conductor triaxial cable having a center conductor, an inner shield, and an outer shield. The center conductor is connected to the RF signal and pre-amp control (RF), the inner shield is connected to ground (GND), and the outer shield is connected to the multi-control bias (diode decoupling control) (MC_BIAS). A 10V power connection may be carried on the same conductor as the RF signal.

In the example embodiment, as explained above with respect to FIG. 5A, coupling electronics portion **203** includes a decoupling circuit, impedance inverter circuit, and pre-amplifier. As illustrated in FIG. 5B, the decoupling circuit includes a decoupling diode **360**. Decoupling diode **360** may be provided with voltage from MC_BIAS, for example, in order to turn decoupling diode **360** on. When on, decoupling diode **360** causes conductor **302-1** to short with conductor **302-2**, thus causing the coil be off-resonance and hence decouple the coil during a transmit operation, for example.

In the example embodiment, the impedance inverter circuit includes a plurality of inductors, including first inductor **370a**, second inductor **370b**, and third inductor **370c**; a plurality of capacitors, including first capacitor **372a**, a second capacitor **372b**, a third capacitor **372c**, and a fourth capacitor **372d**; and a diode **374**. The impedance inverter circuit includes matching circuitry and an input balun. As shown, the input balun is a lattice balun that includes first inductor **370a**, second inductor **370b**, first capacitor **372a**, and second capacitor **372b**. In one example, diode **374** limits the direction of current flow to block RF receive signals from proceeding into decoupling bias branch (MC_BIAS).

In the example embodiment, the pre-amplifier **362** may be a low input impedance pre-amplifier that is optimized for high source impedance by the impedance matching circuitry. The pre-amplifier may have a low noise reflection coefficient, γ , and a low noise resistance, R_n . In one example, the pre-amplifier may have a source reflection coefficient of γ substantially equal to 0.0 and a normalized noise resistance of R_n substantially equal to 0.0 in addition to the low noise figure. However, γ values substantially equal to or less than 0.1 and R_n values substantially equal to or less than 0.2 are also contemplated. With the pre-amplifier having the appropriate γ and R_n values, the pre-amplifier provides a blocking impedance for RF coil **202** while also providing a large noise circle in the context of a Smith Chart. As such, current in RF coil **202** is minimized, the pre-amplifier is effectively noise matched with RF coil **202** output impedance. Having a large noise circle, the pre-amplifier yields an effective SNR over a variety of RF coil impedances while producing a high blocking impedance to RF coil **202**.

In some examples, pre-amplifier **362** may include an impedance transformer that includes a capacitor and an

inductor. The impedance transformer may be configured to alter the impedance of the pre-amplifier to effectively cancel out a reactance of the pre-amplifier, such as capacitance caused by a parasitic capacitance effect. Parasitic capacitance effects can be caused by, for example, a PCB layout of the pre-amplifier or by a gate of the pre-amplifier. Further, such reactance may often increase as the frequency increases. Advantageously, however, configuring the impedance transformer of the pre-amplifier to cancel, or at least minimize, reactance maintains a high impedance (i.e. a blocking impedance) to RF coil 202 and an effective SNR without having a substantial impact on the noise figure of the pre-amplifier. The lattice balun described above may be a non-limiting example of an impedance transformer.

In examples, the pre-amplifier described herein may be a low input pre-amplifier. For example, in some embodiments, a “relatively low” input impedance of the preamplifier is less than approximately 5 ohms at resonance frequency. The coil impedance of RF coil 202 may have any value, which may be dependent on coil loading, coil size, field strength, and/or the like. Examples of the coil impedance of RF coil 202 include, but are not limited to, between approximately 2 ohms and approximately 10 ohms at 1.5 T magnetic field strength. The impedance inverter circuitry is configured to transform the coil impedance of RF coil 202 into a relatively high source impedance. For example, in some embodiments, a “relatively high” source impedance is at least approximately 100 ohms and may be greater than 150 ohms.

The impedance transformer may also provide a blocking impedance to RF coil 202. Transformation of the coil impedance of RF coil 202 to a relative high source impedance may enable the impedance transformer to provide a higher blocking impedance to RF coil 202. Example values for such higher blocking impedances include a blocking impedance of at least 1300 ohms, and at least 1000 ohms.

FIG. 5C shows a cross-sectional view of an example coil loop 201. In the example embodiment, coil loop 201 includes first wire conductor 302-1 and second wire conductor 302-2 surrounded by and encapsulated in dielectric material 303. Each wire conductor may have a suitable cross-sectional shape, herein a circular cross-sectional shape. However, other cross-sectional shapes for the wire conductors are possible, such as elliptical, cylindrical, rectangular, triangular, or hexagonal. The wire conductors may be separated by a suitable distance, and the distance separating the conductors as well as the diameters of the wire conductors may be selected to achieve a desired capacitance. Further, each of first wire conductor 302-1 and second wire conductor 302-2 may be a multi-strand wire conductor, which has a plurality of strands 395, such as a seven conductor stranded wire (e.g., having seven strands), but solid conductors may also be used instead of a stranded wire. A stranded wire may provide more flexibility than solid conductors, at least in some examples.

As appreciated in describing FIG. 5B, the two parallel conductors including the coil loop of an RF coil may each be continuous conductors, or one or both of the conductors may be non-continuous, as illustrated in FIG. 5B. For example, both conductors shown in FIG. 5B may include cuts, resulting in each conductor having two segments. The resulting space between conductor segments may be filled with the dielectric material that encapsulates and surrounds the conductors. The two cuts may be positioned at different locations, e.g., one cut at 135° and the other cut at 225° (relative to where the coil loop interfaces with the coupling electronics). By including discontinuous conductors, the resonance frequency of the coil may be adjusted relative to

a coil that includes continuous conductors. In an example, an RF coil that includes two continuous parallel conductors encapsulated and separated by a dielectric, the resonance frequency may be a smaller, first resonance frequency. If that RF coil instead includes one discontinuous conductor (e.g., where one of the conductors is cut and filled with the dielectric material) and one continuous conductor, with all other parameters (e.g., the conductor wire gauge, loop diameter, spacing between conductors, dielectric material) being the same, the resonance frequency of the RF coil may be a larger, second resonance frequency. In this way, parameters of the coil loop, including conductor wire gauge, loop diameter, spacing between conductors, dielectric material selection and/or thickness, and conductor segment number and lengths, may be adjusted to tune the RF coil to a desired resonance frequency.

FIG. 5D is a cross-sectional view of a wire conductor 302-5D used in coil loop 201 of RF coils 202. Different from coil loops 201 shown in FIG. 5B that include first conductor 302-1, second conductor 302-2, and two driven ends at each end of the conductors, coil loops 201 may include one single wire conductor 302-5D and one driven end 312 at each end of wire conductor 302-5D. When wire conductor 302-5D is used, driven end 312-1 is coupled with an end of wire conductor 302-5D and driven end 312-2 is coupled with the other end of wire conductor 302-5D. Coil loop 201 may form into one turn or a plurality of turns. The resistance of coil loop 201 increases approximately by number of turns, and the loop loss increases approximately by square root of the number of turns, while the body loss increases approximately by the number of turns. As a result, the SNR of coil loop 201 is increased approximately by the square root of the number of turns, such that multiple turns are used to increase the ratio of the body loss over the loop loss, compared with a single turn coil loop. Coil loop 201 forms into a shape of a circle, and may form into other shapes such as a polygon, oval, or irregular shapes. Coil loop 201 defines a conductive pathway along wire conductor 302-5D. Wire conductor 302-5D may be uninterrupted and continuous along an entire length of the coil loop. The coil loop may also include loops of varying gauge of stranded or solid conductor wire, or loops of varying diameters with varying lengths of conductors 302-5D. For example, conductor 302-5D may have no cuts or gaps (no segmented conductors) or one or more cuts or gaps (segmented conductors) at various locations along the conductive pathway. One or more capacitors may be placed at the cuts, gaps, or at the end of the coil loop before driven end 312-2. The capacitance of capacitors may be variable.

In the example embodiment, conductor 302-5D has a suitable cross-sectional shape, such as circular, elliptical, rectangular, triangular, or other shapes that enable conductor 302-5D functions as described herein. Insulating material 303 surrounds conductors 302-5D. Dielectric material 303 may be rubber, plastic, or any other dielectric material. conductor 302-5D includes one or a plurality of strands 395. For example, conductor 302-5D is a single-strand wire conductor. Alternatively, conductor 302-5D is a multi-strand wire conductor having a plurality of strands 395, where an individual strand 395 may be surrounded by insulating material or not surrounded by insulating material. Individual strands 395 may be twisted upon each other or may be parallel to each other, along the length of strand 395. In one example, wire conductor 302-5D includes 19 strands that are 36 AWG each for an overall thickness of 24 AWG, and the cross section of wire conductor 302-5D has a diameter of 0.025 inches (0.06 cm). A coil loop 201 including multi-

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strand conductors **302-5D** has a higher penetration depth and a higher SNR than a coil loop **201** of the same diameter that includes a single-strand conductor. Therefore, the size of coil loop **201** may be reduced by including multi-strand wire conductors **302-5D** for the same penetration depth, and consequently an increased number of RF coils **202** may be included in a coil array.

The RF coils **202** presented above with respect to FIG. **5A** may be used in order to receive MR signals during an MR imaging session. As such, the RF coils of FIG. **5A** are configured to be coupled to downstream components of MR system **10**. RF coils **202** of FIG. **5A** may be present in an array of RF coils having various configurations.

FIGS. **6A** and **6B** shows example wiring harness **1306**. In the example embodiment, wiring harness **1306** includes a plurality of module interfacing cables **602** extending from a module interfacing connector **603**. Module interfacing cables **602** may be used to connect the RF coil assembly **1300** to other components of the MR system **10** such as the RF system **30** through a coil assembly interfacing connector **608**. The module interfacing cable **602** may include a baluns **607** or contiguous/continuous distributed baluns (not shown).

In the example embodiment, coupling electronics portions **203** or part of coupling electronics portions **203** are assembled in electronics box **1414**. Electronics box **1414** includes an outlet **604** configured to electrically couple with module interfacing connector **603**, which is electrically coupled with coil assembly interfacing connector **608**. Coil assembly interfacing connector **608** further couples to other components of MR system **10** such as RF system **30**. For example, when RF coil assembly **1300** is in use, coil assembly interfacing connector **608** is plugged in a coil interface (not shown) or port, coupling RF coil assembly **1300** to the rest of the MR system **10**, such as RF system **30**.

In the example embodiment, RF coil assembly **1300** further includes a control circuitry. The control circuitry is the MC_BIAS for switching RF coils between receive and decoupled modes (see FIG. **5B**). Elements of the control circuitry may be incorporated in coupling electronics portion **203**, module interfacing connector **603**, and/or coil assembly interfacing connector **608**.

In the example embodiment, RF coil assembly **1300** is flexible without increased stress on coil loops **201** and other components of RF coil assembly **1300**. As used herein, an RF coil assembly is flexible when the RF coil assembly may be flexed or bent to change the shape of the RF coil assembly. The RF coils **202** described above are configured to maintain the performance while the RF coils are flexed or bent.

In some embodiments, modules **1302** wirelessly communicate with coil assembly interfacing connector **608** or RF system **30**. Module interfacing cables **602** are removed. Outlet **604** of module **1302** may include a wireless communication connector, such as a radio wave transmitter and receiver, configured to wirelessly communicate with coil assembly interfacing connector **608** or RF system **30**. Signals are transmitted between modules **1302** and coil assembly interfacing connector **608** and further to RF system **30**, or directly between modules **1302** and RF system **30**.

In operation, to assembly an RF coil assembly **1300**, modules **1302** are coupled with one another. Modules **1302** may be coupled with one another via coupler **1408**. Electronics boxes **1414** of coupled modules **1302** may be positioned on the same side of RF coil assembly (see FIG. **6A**). Alternatively, electronics boxes **1414** may face any direction. Module interfacing cables **602** are coupled with outlets

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604 of modules **1302**. Module interfacing cables **602** may be plugged into outlets **604**. In some embodiments, wiring harness **1306** is coupled with modules **1302** by powering on outlet **604** to provide wireless communication. In other embodiments, wiring harness **1306** is eliminated and modules **1302** communicate wirelessly with RF system **30**. During scanning, RF coil assembly **1300** is placed on the subject. Electronics boxes **1414** may face away from the subject, to increase patient comfort. Coil assembly interfacing connector **608** is plugged into a port of MR system **10**.

As described above, in MR, signals are acquired by an RF coil. Therefore, an RF coil plays a major role in image quality, such as SNR or image distortion, of images acquired by the MR system. RF coils are desirable to be flexible such that RF coils conform to and are proximate to the anatomy of the subject. RF coil assemblies **1300** described herein accomplish this goal.

FIGS. **7A-7C** show another known modular RF coil assembly **1200-7**. Like the other known RF coil assembly **1200-2**, known RF coil assembly **1200-7** is constructed with rigid coil loops **1204** of copper strips. Coil loops **1204** are coupled with neighboring coil loops at a predefined configuration such that a coil loop **1204** is critically overlapped with its neighboring coil loops. To maintain the formation, RF coil assembly **1200-7** is embedded in a patient table **702** of the MR system (FIGS. **7A** and **7B**). Modules **1202** are connected to a common strip line **704** embedded in table **702**.

FIGS. **8A-8D** show more example embodiments of RF coil assembly **1300-8**. In the example embodiments, module **1302** of RF coil assembly **1300** includes RF coils in arrays. RF coils may be RF coils **202** described above. In some embodiment, coil loops of RF coil assembly **1300** may be fabricated with metal strips. RF coils may be form in arrays. RF coil assembly **1300-8** may include a strip line **804** through modules **1302** and configured to transmit signals to and from modules **1302**. Modules **1302** may be pluggable into strip line **804**.

Compared to known RF coil assembly **1200-7**, modules **1302** of RF coil assembly **1300** may have different dimensions and may include different numbers of RF coils therein. In known RF coil assembly **1200-7**, each module **1202** has the same number of RF coils, such as 1x4 or 2x4 (see FIG. **7B**). In contrast, modules **1302** of RF coil assembly **1300** have different dimensions. For example, module **1302-s** is smaller than module **1302-1** and may include a fewer number of RF coils. Module **1302-s** may include 1x4 RF coils while module **1302-1** may include 4x4 RF coils. Different sizes of modules **1302** may be included in constructing RF coil assembly **1300** such that different sizes of field of view FOV may be used to scan different areas of anatomies of subject **1304**. For example, to scan a head **806**, a desired FOV is the size of head **806** such as 20 cm. A small module **1302-s** may be placed at an end **808** of RF coil assembly **1300** (see FIGS. **8B-8D**). Similarly, if images of feet **810** or legs **812** are to be acquired, small modules **1302-s** are placed at corresponding sections of RF coil assembly **1300** (see FIGS. **8C** and **8D**). If images of a large FOV of subject **1304** are desired, large modules **1302-1** are placed at corresponding part of subject **1304**.

Compared to known RF coil assembly **1200-2**, **1200-7**, RF coil assembly **1300** is advantageous for being serviceable. For example, in known RF coil assembly **1200**, modules **1202** need to be placed at a predefined configuration and coupling of modules **1202** are complicated, nonfunctioning of one module **1202** renders nonfunctioning of the entire RF coil assembly **1200**. A clinician may not simply replace the

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nonfunctioning module with a spare module because of the strict predefined configuration and complicated coupling. Further, because known RF coil assembly **1200-7** is embedded in table **702**, if a module **1302** does not function and needs to be repaired, a clinician could not simply replace the nonfunctioning module **1202** and return RF coil assembly **1200** to operation. A service is needed to take the table apart, remove the nonfunctioning module **1202**, replace with a new module, and return RF coil assembly **1200** to functioning state.

In contrast, RF coil assembly **1300** is serviceable. a clinician may simply replace the nonfunctioning module with a spare module stored in the scanner room and return RF coil assembly to operation. Further, modules **1302** are individually controlled and connected with MR system **10**. If a spare module **1302** is not available, the rest of RF coil assembly **1300** still functions, without interruption of operation.

When RF coil assembly **1300** includes RF coil **202** having coil loop **201** formed by wire conductor **302**, because of the flexibility of wire conductor **302** and outer enclosure, modules **1302** are flexible and deformable and configured to deform into shapes congruent with the shapes of subject **1304** such that RF coils **202** are placed close to anatomies of interest, thereby increasing SNR of the signals received by RF coil assembly **1300** and efficiency of RF coil assembly **1300**. Further, wire conductor **302**, RF coil **202**, and outer enclosure **1420** are lighter than traditional RF coil assembly. The weight of RF coil assemblies described herein is significantly reduced. For example, the known RF coil assembly **1200-2** may weigh 1 lb or more for one module, which may be unsuitable to be placed on a child, an elderly, or a sick subject. In contrast, RF coil assemblies described herein weighs approximately 0.5 g/cm², suitable to be placed on any subject, thereby increasing subject comfort.

Because modules **1302** in RF coil assembly **1300** may be manufactured using the same material, the material cost may be reduced. With modular RF coil assembly **1300**, the number of different components in an RF coil assembly is greatly reduced. Therefore, the number of suppliers is reduced, increasing the quality of an RF coil assembly and reducing the repair cost. Further, because the process of building and tuning the RF coil assemblies becomes the same for the same modules, the manufacturing process is controlled and automated, and waste from manufacturing is controlled, thereby reducing cost of labor and overhead. In addition, because the modules are replaceable and serviceable in the field rather than incurring down time from waiting for replacement of the entire RF coil assembly in known RF coil assemblies, the repair cost as well as shipping costs are reduced.

Because modules **1302** of RF coil assembly **1300** may be arranged in any orientation, and overlapped or underlapped at any distance, RF coil assembly **1300** may be configured as an anterior RF coil assembly, a posterior RF coil assembly, a head RF coil assembly, a whole body assembly, a musculoskeletal (MSK) RF coil assembly, or any combination thereof. RF coil assembly **1300** may be configured as having two or more types of coils such as a head RF coil assembly and an anterior RF coil assembly by including two groups of modules **1302** in RF coil assembly **1300** with groups underlapped from one another. As a result, flexibility in constructing RF coil assembly **1300** is significantly increased.

RF coil assemblies described herein are also advantageous over a mere inclusion of multiple RF coil assemblies in MR system **10**. MR system **10** has a limited number of

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ports for RF coils to be plugged in. MR system **10** typically has four ports. Therefore, the number of RF coil assemblies are conventional limited to four. In MR systems, because of the interactions between RF and magnetic field, care is needed in placement of cables of RF coil assemblies to avoid cross talk or signal loss. Cables need to long for the RF coil assemblies to be plugged in the ports. The long cables are also cumbersome in set-up with a subject because the long cables should be placed away from the subject to provide comfort. In contrast, RF coil assemblies described herein need one port for one RF coil assembly, increasing the flexibility of MR system **10** and the number of RF coil assemblies in MR system **10**. Because modules **1302** are coupled to wiring harness **1306**, problems associated with multiple long cables are obviated.

RF coil assemblies described herein are also advantageous in providing flexibility in system setup. Modules to be included in an RF coil assembly may be customized to fit imaging needs. For example, modules of relatively small dimensions may be used to image a small FOV, such as the head. Modules of relatively big dimensions may be used to image a large FOV, such as the spine. Therefore, when imaging a small FOV, a large RF coil assembly is not placed on the subject, thereby increasing patient comfort, image quality, and SNR because small coil loops are closer to anatomies of interest than large coil loops. RF coil assemblies provides the capabilities of imaging a large FOV without the need of using a separate RF coil assembly. Instead, a large module or large modules are included in one RF coil assembly. Further, the changes may be performed by a clinician because a module may be plugged in or removed from an RF coil assembly.

FIG. **9** is a flow chart of an example method **900** of assembling an RF coil assembly. The RF coil assembly may be RF coil assemblies **1300** described above. Method **900** includes providing **902** a first module and a second module. Modules may be modules **1302** described above. Method **900** further includes coupling **904** the first module with the second module. Method **900** also includes coupling **906** the modules with the RF system of the MR system. Coupled modules **1302** may be coupled to RF system **30** via a wiring harness **1306**. In some embodiment, coupled modules **1302** is coupled with common strip line **804**, which is configured to couple with RF system **30**. In other embodiments, coupled modules **1302** are coupled with RF system **30** through wireless communication.

At least one technical effect of the systems and methods described herein includes (a) modular RF coil assemblies; (b) a wiring harness for coupling modules; (c) pluggable connections between modules and the wiring harness; (d) flexibility in construction of an RF coil assembly; (e) a serviceable RF coil assembly, and (d) a simplified assembling process.

Example embodiments of assemblies, systems, and methods of RF coil assemblies are described above in detail. The systems and methods are not limited to the specific embodiments described herein but, rather, components of the systems and/or operations of the methods may be utilized independently and separately from other components and/or operations described herein. Further, the described components and/or operations may also be defined in, or used in combination with, other systems, methods, and/or devices, and are not limited to practice with only the systems described herein.

Although specific features of various embodiments of the invention may be shown in some drawings and not in others, this is for convenience only. In accordance with the prin-

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ciples of the invention, any feature of a drawing may be referenced and/or claimed in combination with any feature of any other drawing.

This written description uses examples to disclose the invention, including the best mode, and to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

What is claimed is:

1. A radio frequency (RF) coil assembly for a magnetic resonance (MR) system, comprising:

one or more modules, a module of the one or more modules comprising:

one or more RF coils, each RF coil comprising:

a coil loop that comprises a wire conductor, the wire conductor formed into the coil loop; and

a coupling electronics portion coupled with the coil loop; and

an outer enclosure housing the one or more RF coils and separating one of the one or more modules from another one of the one or more modules; and

an outlet coupled with coupling electronics portions of the one or more RF coils; and

a wiring harness comprising a plurality of module interfacing cables and a coil assembly interfacing connector, the plurality of module interfacing cables bundled and extending from the coil assembly interfacing connector, the plurality of module interfacing cables comprising one or more baluns, one of the plurality of module interfacing cables configured to be coupled with an outlet of the one of the one or more modules, another one of the plurality of module interfacing cables configured to be coupled with an outlet of another one of the one or more modules.

2. The RF coil assembly of claim 1, wherein the one of the plurality of module interfacing cables of the wiring harness is pluggable in the outlet of the one of the one or more modules.

3. The RF coil assembly of claim 1, wherein one of the one or more modules overlaps with another one of the one or more modules at an arbitrary orientation.

4. The RF coil assembly of claim 1, wherein the wiring harness is configured to be electrically coupled to a port of the MR system.

5. The RF coil assembly of claim 1, wherein a first module of the one or more modules overlaps with a second module of the one or more modules and underlaps with a third module of the one or more modules.

6. The RF coil assembly of claim 1, wherein the module further comprises:

a coupler configured to couple with a coupler of another module.

7. The RF coil assembly of claim 6, wherein a first module is critically overlapped with a second module when a coupler of the first module is aligned with a coupler of the second module.

8. The RF coil assembly of claim 1, wherein the module further comprises:

an electronics box sized to house coupling electronics portions of the one or more RF coils and the outlet.

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9. The RF coil assembly of claim 8,

wherein the electronics box is positioned at an exterior of the outer enclosure.

10. A radio frequency (RF) coil assembly for a magnetic resonance (MR) system, comprising:

a first module comprising:

a first number of RF coils, each RF coil comprising a coil loop; and

a first outer enclosure housing coil loops of the first number of RF coils;

a second module comprising:

a second number of RF coils, the second number being different from the first number; and

a second outer enclosure housing coil loops of the second number of RF coils,

wherein the first module and the second module are configured to be installed on a table of the MR system; and

a strip line configured to be installed in the table of the MR system, the strip line extending through the first module and the second module and configured to individually couple with the first module and the second module.

11. The RF coil assembly of claim 10, wherein the coil loop further comprises a wire conductor, the wire conductor formed into the coil loop, and the RF coil further comprises: a coupling electronics portion coupled with the coil loop.

12. The RF coil assembly of claim 11, wherein the first outer enclosure and the second outer enclosure are configured to deform into shapes congruent with a shape of a subject of the MR system.

13. The RF coil assembly of claim 10, wherein at least one of the first module or the second module is pluggable into the strip line.

14. A module of a radio frequency (RF) coil assembly for a magnetic resonance (MR) system, comprising:

one or more RF coils, each RF coil comprising:

a coil loop that comprises a wire conductor, the wire conductor formed into the coil loop; and

a coupling electronics portion coupled with the coil loop;

an outer enclosure housing the one or more RF coils and separating the module from another module; and

an outlet extending through the outer enclosure and coupled with coupling electronics portions of the one or more RF coils, the outlet being exposed and configured to individually establish wired communication between the module and the MR system via a wiring harness, wherein the wiring harness includes a plurality of module interfacing cables and a coil assembly interfacing connector, the plurality of module interfacing cables bundled and extending from the coil assembly interfacing connector, one of the plurality of module interfacing cables configured to be coupled with the outlet, another one of the plurality of module interfacing cables configured to be coupled with an outlet of another module.

15. The module of claim 14, wherein the outlet is complementary to the one of the plurality of module interfacing cables of the wiring harness, thereby being pluggable in coupling with the one of the plurality of module interfacing cables of the wiring harness.

16. The module of claim 14, wherein the module further comprises:

a coupler configured to couple with a coupler of another module.

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17. The module of claim **16**, wherein the module is critically overlapped with another module when the coupler of the module is aligned with the coupler of another module.

18. The module of claim **14**, wherein the module further comprises:

an electronics box sized to house coupling electronics portions of the one or more RF coils and the outlet.

19. The module of claim **18**, wherein the module further comprises:

an outer enclosure housing coil loops of the one or more RF coils,

wherein the electronics box is positioned at an exterior of the outer enclosure.

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